Atomic Spectroscopy

Review

A Review of Development in the Research of Self-Absorption on Laser-Induced Breakdown Spectroscopy

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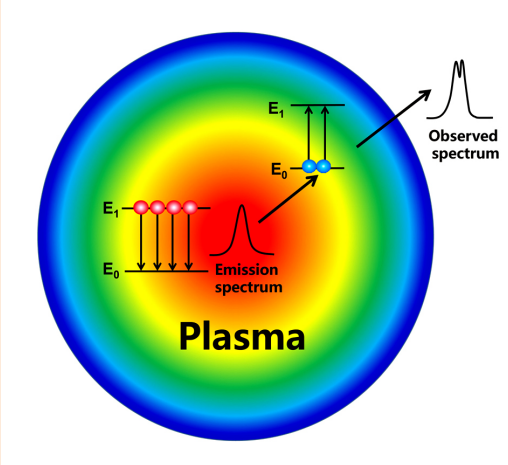
ABSTRACT: Laser-induced breakdown spectroscopy (LIBS) has been successfully applied to various fields since its inception in the late twenty century. However, there are still challenges to overcome in order to enhance its performance in analytical experiments. These challenges include three major challenges: matrix effect, spectral interference, and self-absorption effect. This review specifically focuses on investigating the self-absorption effect, which disrupts the linear relationship between spectral line intensity and element concentration, thus affecting the the accuracy and sensitivity of LIBS analysis. The review explores different approaches to evaluate the extent of self-absorption, taking into account the evolutionary mechanism of this effect. Additionally, various methods for correcting spectra have been proposed, showing potential in significantly reducing self-absorption. Notably, the review emphasizes proactive measures to create non-self-absorbed conditions as the most effective approach to mitigate the self-absorption effect. Finally, the review presents a schematic description of exploiting self-absorption and highlights its promising prospects for future applications of LIBS.



INTRODUCTION

In 1962, a publication discussed the generation of plasma through the utilization of laser pulses.[1] Two decades later, Loree *et al.*[2] utilized the laser-induced plasma to analyze element concentrations, marking significant milestone in the advent of LIBS. The principle involves ablating and exciting the substance in the sample, typically a solid material, using a spectrometer to determine the elemental composition of the sample for both qualitative and quantitative analysis.

After more than 60 years of development, LIBS has gained widespread recognition for its qualitative analytical capabilities, largely due to advancements in laser technology. Compared to other spectroscopy approaches,[3-5] LIBS offers several distinct advantages, including (i) a simple and quick procedure that does not require sample preparation; (ii) simultaneous multi-elemental analysis;[6] (iii) competence for remote detection;[7] (iv) in-situ analysis;[8-10] (v) validity for a broad range of analytes,[10-16] among others. Currently, it is widely used in atmospheric particle detection,[17] corrosion detection,[18] environmental and agricultural,[19] food crop detection,[20] and medical diagnosis.[21, 22]

However, the LIBS technique faces several critical challenges due to the complex of the spatial-temporal evolution of plasma. These challenges include matrix effects,[23] spectral interference,[24-26] and self-absorption[27] which hinders the further development and

application of this technology. Compared to traditional elemental analysis technologies such as Inductively Coupled Plasma-Mass Spectrometry (ICP-MS) ,[28] X-ray Fluorescence Spectrometry(XRF) ,[29] and Electron Probe Micro-Analysis (EPMA) ,[30] LIBS still exhibits lower accuracy in quantitative analysis, primarily due to the self-absorption effect.[31] The self-absorption effect occurs when emission lines originating from the center of the plasma experience absorption as they propagate toward the outer regions, resulting in a reduction in their intensities. This effect disrupts the linearity between element concentration and line intensity, thus compromising the precision of quantitative analysis. Therefore, addressing the challenge of self-absorption is crucial for improving the accuracy of quantitative analysis. This review aims to provide a comprehensive of relevant literature, offering readers in this field a thorough understanding of the implications, evaluation, reduction, and exploitation of the self-absorption effect in LIBS.

Numerous scholars have conducted research on self-absorption in LIBS, covering various aspects including the physical mechanisms, the evaluation methodologies, methods for correction and suppression, and the exploitation approaches.[32-34] This review provides a comprehensive summary of self-absorption aspects in LIBS, In section 2, we present typical spectra exhibiting symptoms of self-absorption and discuss the underlying principle of this effect. Also, evolution behaviors with some crucial experimental parameters have been discussed on the practical demands. For instance, laser parameters such as the laser wavelength and intensity often have a complex impact on self-absorption, interacting with other parameters like background gas pressure. Various approaches have been proposed to evaluate the degree of self-absorption. Subsequent sections will introduce several correction approaches to readers based on our understanding of the mechanism. In this study, we focus more on the suppression methods, particularly the utilization of assisted excitation techniques. Additionally, we explore a novel perspective of exploiting self-absorption in a reverse manner to obtain valuable information.

MECHANISM

In 1923, Kimura *et al.* [35]observed the self-absorption effect in the spectral lines of H_α and H_β . Subsequently, Wood *et al.* [36] concluded from repeated experiments that the self-absorption effect is caused by the absorbing layer within the plasma. Generally, in homogeneous plasma, the effect of self-absorption on the spectrum is weak, and it is challenging to determine the presence of self-absorption based on the morphology of the spectrum. Typically, self-absorption is more severe in the center of the spectral line than in the wings. When the center of the self-absorption spectral line becomes concave (refer to Fig. 1), it can

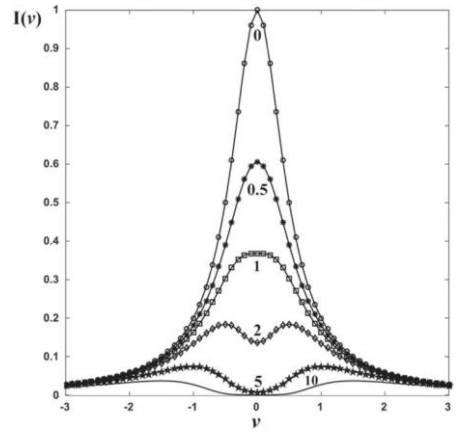


Fig. 1 Examples of the spectral profile affected by self-absorption.

lead to the formation of double peaks known as "self-reversal". This phenomenon disrupts the linear mapping relationship between the intensity of the spectral line and the concentration of the measured element, resulting in a reduction in spectral line intensity with further increases in concentration, and in some cases, it can even drop to zero. Such effects have a significant impact on the accuracy of quantitative analysis in LIBS. Additionally, due to the inhomogeneity of the laser-induced plasma, the concavity of the spectral lines may deviates from the center and exhibit asymmetric shapes.[37]

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According to the classical spectroscopic theory of LIBS, under ideal conditions, when the plasma satisfies local thermodynamic equilibrium (LTE) and is optically thin, the intensity of the emission spectrum from laser-induced plasma exhibits a linear correlation with the elemental concentration in the sample. However, in practical applications, the laser-induced plasma often exists in an optically thick state, and the spatial distribution of temperature within the plasma becomes inhomogeneous. Specifically, the center region of the plasma has a higher temperature, while the outer region has a lower temperature. Additionally, the majority of particles in the plasma's low energy level are in the ground state. Consequently, when the high-temperature region in the plasma's center emits energy outward, it gets absorbed by similar particles in the lower energy level present in the outer layer. This absorption leads to a reduction in the

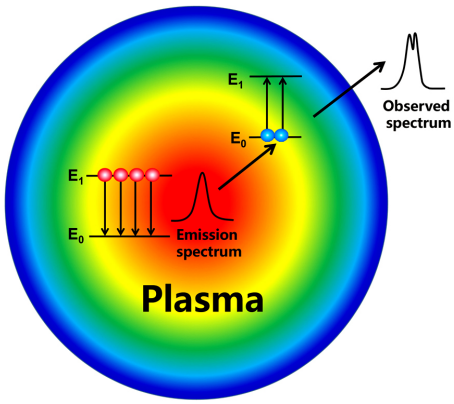

Fig. 2 A schematic illustration of the mechanism of the self-absorption effect in LIBS. The observed spectrum is under the concerted force of emission and absorption spectrum.

intensity of the emission spectrum.

Figure 2 illustrates the generation mechanism of the self-absorption effect. When particles at the center of the plasma that are in a high energy level spontaneously radiate and transition to a low energy level, they emit photons outward. When the radiated photons pass through the low-temperature region of the outer layer, a large number of particles in the lower energy levels accumulate inside the region. When these particles meet the energy matching conditions for energy level transition, they will absorb the photons radiated from the higher energy level and transition from the lower energy level to the energy level, a phenomenon known as stimulated absorption. This absorption phenomenon affects the intensity of the radiation spectrum, resulting in a weakening of the observed spectral intensity, thus giving rise to the self-absorption effect.

EVALUATION

The self-absorption effect can exhibit a spectrum of effects on experimental diagnosis, ranging from negligible to extremely pronounced. It is crucial to assess the extent of this effect before further application of LIBS to guarantee precise outcomes. It is often difficult to judge whether a self-absorption effect exists from the shape of spectral lines, so methods for evaluating the severity of the self-absorption effect need to be established. Several methods for evaluating self-absorption effects are described below.

Self-absorption (SA) coefficient parameter. El Sherbini *et al.*[38] initially proposed the self-absorption (SA) coefficient method based on the symptomatic spectral profile. This approach involves the selection of specific spectral parameters to determine the self-absorption coefficient. The core principle lies in comparing a particular spectral line with and without self-absorption effects. Various definitions of the SA coefficient willed introduced, along with its application in different experimental situations.

The most commonly method for determining the SA coefficient involves utilizing the peak ratio. The SA coefficient parameter is defined as the ratio of the observed peak value of line intensity to the peak values in the absence of self-absorption. Therefore, the SA coefficient can be expressed as[38, 39]:

$$SA = \frac{I(\lambda_0)}{I_0(\lambda_0)} = \frac{1 - \exp[-\kappa(\lambda_0)l]}{\kappa(\lambda_0)l} \quad (1)$$

Here, λ_0 is the center wavelength of the spectral line, $\kappa(\lambda_0)l$ is the optical depth, and $I_0(\lambda_0)$ is the spectral intensity of the line without self-absorption. Generally, in LIBS, the Gaussian contribution to the spectral line width is neglected. Thus, the optical depth and the SA coefficient parameter can be expressed as follows:

$$SA = \frac{1-\exp[-\kappa(\lambda_0)l]}{\kappa(\lambda_0)l} = \Delta\lambda_0 \frac{1-\exp(\frac{-K}{\Delta\lambda_0})}{K} \quad (2a)$$

$$\kappa(\lambda)l \approx K \frac{\Delta\lambda_0}{4(\lambda-\lambda_0)^2+\Delta\lambda_0^2} \quad (2b)$$

where $K = \frac{2e^2}{mc^2} n_i f \Delta\lambda_0^2 l$, e and m are the charge and the mass of the electron, respectively; f is the oscillator strength, and $\Delta\lambda_0$ is the line full width at half maximum (FWHM). Apparently, when the plasma is in an optically thin state, i.e., there is no self-absorption effect in the spectral lines, the value of SA is 1. When the self-absorption effect is enhanced, the value of SA decreases until the plasma is in an optically thick state, at which time the value of SA is infinitely close to 0. Moreover, the SA coefficient can be expressed in the reciprocal form[40]:

$$F_{max}^{L} = \frac{I_0^{thin}}{I_0^{thick}} = \frac{\kappa(\lambda_0)l}{1 - \exp[-\kappa(\lambda_0)l]} \quad (3)$$

It should be noted that the SA parameters as described in Eq (3) and subsequent sections are assumed to be applicable within a homogeneous plasma circumstance. The expression of SA can also be represented in relation to the integrated line intensity[38]:

$$(SA)^{\beta} = \frac{I_{int}^{thick}}{I_{int}^{thin}} \quad (4)$$

Where β = 0.46 in general. As shown above, the SA coefficient is expressed as the ratio of the integrated line intensities of the spectral lines when the plasma is optically thick and thin, respectively.

Besides, Di Rocco *et al.*[41] introduced a universal curve of τ_0 (the optical depth as usual) for any spectral lines, which can be expressed as:

$$U(\tau_0) = \frac{I(\tau_0)FWHM(\tau_0)}{S(\tau_0)} \quad (5)$$

where $S(\tau_0)$ is related to the integrated line intensity. According to the literature, the typical value of the universal curve is around $2/\pi$ for optically thin lines. Limitations of this method include its inability to differentiate between pure Lorentzian and self-absorption, especially for case where $\tau_0 \geq 1$, as well as its sensitivity to the spectral noise. Nevertheless, this approach can be employed to ascertain the optical depth even in the absence of prior information regarding the plasma parameters.

By the definition of the half-height width of the spectral line (i.e., the value of $I(\lambda)$ should be equal to $I(\lambda_0)/2$, when $\lambda = \lambda_0 \pm \Delta\lambda/2$), therefore, the new SA parameter, defined as the ratio of the observed $\Delta\lambda$ over the corresponding non affected $\Delta\lambda_0$, is expressed as:

$$(SA)^{\alpha} = \frac{\Delta\lambda^{thick}}{\Delta\lambda^{thin}} \quad (6)$$

Where $\Delta\lambda^{thick}$ is the observed spectral line FWHM with self-absorption; $\Delta\lambda^{thin}$ is the spectral line FWHM without self-absorption effects; $\alpha = 0.54$ in the case of the Lorentzian profile. By the way, the ratio $\beta/\alpha = 0.82$ calculated by Praher *et al.*[42] is very close to the 0.85 derived in the original literature. , When $\Delta\lambda_0 \approx 2w_s(T)n_e$, the collisions of atoms and ions are negligible, and the Stark broadening parameter can be determined, giving:[38]

$$\Delta\lambda = \Delta\lambda_0(SA)^{\alpha} \approx 2w_s(T)n_e(SA)^{\alpha} \quad (7)$$

By the way, Konjević *et al.* [43] also studied the correlation between self-absorption and the Stark broadening, but they needed to determine the electron density and the Stark broadening parameter of the lines. However, it may be not capable this method is not applicable in the presence of extreme self-absorption condition, due to the line shape cannot be fitted with the Lorentzian function. While determining the SA parameter is simple, calculating the optical depth in Eq. (1) requires knowledge of the number density of the emitting species, which is often unavailable experimentally in general.

In the context of multispectral analysis, previous methodologies have primarily focused on the spectral parameters derived from a single spectrum. To enhance the versatility of the SA coefficient method across diverse experimental settings, Bredice *et al.* [44] proposed an innovative approach that involves comparing two or more LIBS spectra obtained from the same analyte at different delay times.

According to the classical Boltzmann plots, in the approximation

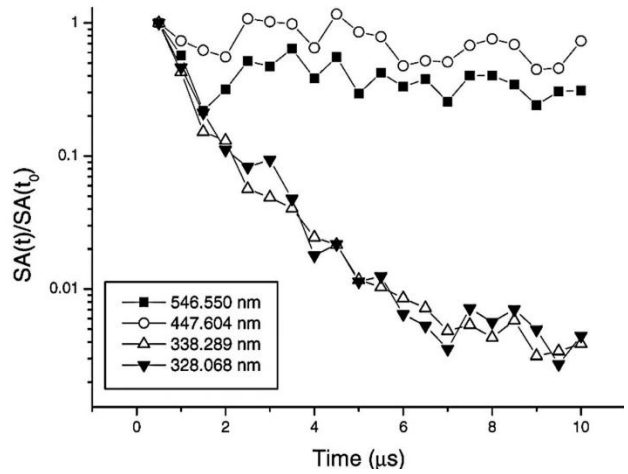

Fig. 3 $SA(t)/SA(t_0)$ via delay time for the Ag I line.

of LTE and without self-absorption efftecs, a straight line with the slope of $-1/(K_B\,T)$ in the Boltzmann plot is, from which the value of plasma temperature can be derived. The value of the slope is given by:

$$\tan\alpha = \frac{\ln\left[\frac{I_{02}(T_a)}{I_{02}(T_b)}\right] - \ln\left[\frac{I_{01}(T_a)}{I_{01}(T_b)}\right]}{E_{k2} - E_{k1}} = \frac{1}{K_B}\left(\frac{1}{T_b} - \frac{1}{T_a}\right) \quad (8)$$

where subscripts a and b stand for two different spectrums; E_{k1}, E_{k2} represent the upper-level energy; K_B is the Boltzmann constant. In scenarios where E_{k1} approaches 0 and $E_{k2} = E_k$, adjustments to the logarithm of the ratio necessary to account for the SA coefficients in the spectral line:

$$\frac{\ln\left\{\frac{I(T_a)}{I(T_b)}\left[\frac{(SA)(T_a)}{(SA)(T_b)}\right]^{\beta}\right\}}{E_k} = \frac{1}{K}\left(\frac{1}{T_b} - \frac{1}{T_a}\right) \quad (9)$$

Therefore, when plasma temperature T_a becomes high enough then it holds the state of optically thin, the SA parameter can be calculated straightforwardly by:

$$\frac{\ln\left\{\frac{I(T_a)}{I(T_b)}[(SA)(T_b)]^{\beta}\right\}}{E_k} \approx \frac{1}{K}\left(\frac{1}{T_b} - \frac{1}{T_a}\right) \quad (10)$$

Moreover, if f at least one non-self-absorbed line is known for the same species, the Eq. (10) can then be simplified by introducing this reference line I_R:

$$\frac{\ln\left[\frac{I_R(T_a)}{I_R(T_b)}\right]}{E_{kR}} = \frac{1}{K}\left(\frac{1}{T_b} - \frac{1}{T_a}\right) \quad (11)$$

D. M. D. Pace *et al*[45] studied the temporal evolution of the SA parameter of Ag I lines (328.608, 338.289, 447.604 and 546.550 nm) with different delay times, as depicted in Fig. 3. The authors suggested that this approach might be applied in experiments with

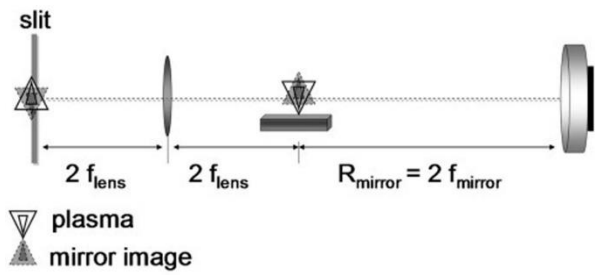

Fig. 4 A schematic experimental arrangement of the duplicating mirrors.

low spectral resolution spectrometers. What's neat is that determining the SA parameter doesn't hinge on plasma parameters or the line spectral coefficients like the transition probability, so this approach is handy for trials using spectrometers with limited spectral resolution. This means it's a breeze to calculate the SA parameters even when the instrumental broadening order matches the FWHM.

Duplicating mirror. This method sets itself apart from those mentioned above. Gouy[46] introduced the concept of the duplicating mirror, where an extra spherical mirror is used to replicate the optical path length of the radiation through the plasma source. The schematic experimental configuration is shown in Fig. 4. Moon *et al*.[47] were the trailblazers in implementing this technique for LIBS, in 2009. They ingeniously positioned a mirror behind the plasma at double the focal length to duplicate the optical path length. By analyzing the emission data with and without the mirror, researchers can ascertain the presence of self-absorption. When self-absorption effect can be neglected, the elected line intensity will be doubled as strong. However, if influenced by self-absorption, the line intensity will fall short of doubling. The radiation captured with and without the mirror are given by[48]:

$$I_1 = B(1 - e^{-\tau(\lambda)l}) \tag{12}$$

$$I_2 = I_1(1 + Ge^{-\tau(\lambda)l}) \tag{13}$$

Where the parameter G represents the loss factor accounting for reflection and absorption caused by mirror. The subscripts 1 and 2 differentiate between scenarios without and with the mirror. The relationship between the two line intensities, denoted as R_λ, serves as a gauge for self-absorption, it can be expressed as:

$$R_\lambda = \frac{I_2}{I_1} = 1 + Ge^{-\tau(\lambda)l} \tag{14}$$

When it comes to continuum radiation, R_λ is equal to 1+G ($\tau(\lambda) = 0$). Therefore, by combination Eq. (14) with the corresponding continuum emission value, we can cleverly deduce the optical depth:

$$\tau(\lambda)l = \ln(R_C - 1) - \ln(R_\lambda - 1) \tag{15}$$

Where the index C represents the continuum radiation. Thus, if the line's intensity, with the help of the reflecting mirror, can be approximated as continuum radiation, then the line is deemed to be slightly self-absorbed. In conclusion, the degree of the self-absorption effect can be easily gauged through the duplicating mirror method. Nevertheless, there have been observations of reduced sensitivity in low-intensity situations.

Curve of growth (COG). The COG showcases how the concentration of elements concentration to line intensity in a double-logarithmic plot. This nifty concept was first cooked up by Ladenburg and Reiche in 1910, and then Mitchell jazzed it up even more to dive into analyzing those self-absorbed resonance lines[49]. In 1999, Gornushkin *et al*.[50] applied this concept to investigate the evolutionary mechanism of plasma in LIBS. They explored how the experimental growth curve is influenced by the parameter α, which reflects the balance between collisions and Doppler broadening. When α exceeds 1, the curve exhibits two asymptotes with slopes — one at 1 and the other at1/2. As for the intriguing relationship between line intensity and species concentration x at varying density, Aragon *et al*.[51] derived the following results:

$$I \cong I_p A x \tag{16}$$

$$I \cong I_p [2A(\Delta\lambda_L)]^{\frac{1}{2}} x^{\frac{1}{2}} \tag{17}$$

Where the Plank distribution represents black body emission, $\Delta\lambda_L$ is the Lorentzian line width. The equation $A = \kappa N' l/x$ and A relates to the absorption coefficient, and N' represents the normalized number density. Calculating the abscissa of the point where these two asymptotes intersect is a simple process:

$$x = \frac{2(\Delta\lambda_L)}{A} \tag{18}$$

The concentration at which the transition occurs from an optically thin to a thick plasma occurs can be regarded as a critical point. Many research teams have employed the COG method in LIBS studies to analyze element composition and assess plasma characterize. For instance, Aguilera *et al*.[52] investigated the temporal evolution and spatial inhomogeneity of plasma. Y. Tang *et al.*[53] recently proposed a method for estimating SA by utilizing exponential fitting of the calibration curve corresponding to a particular emission line under study. They introduced a parameter α, which satisfies the following relation:

$$\alpha C = \kappa(\lambda_0) l \tag{19}$$

Where C is the element concentration, while the term on the opposite side represents the optical thickness. After figuring out this parameter through fitting, you can evaluate the extent of the self-absorption effect as follows:

$$SA = \frac{1 - e^{-\alpha C}}{\alpha C} \tag{20}$$

The COG method presents a level of complexity when studying laser-induced plasma and constructing calibration curves, both in theoretical understanding and practical application. Nonetheless, the spectrochemical analysis using LIBS can be significantly improved. It is feasible to determine the limits of the linear spectral region and choose suitable lines for analysis.

CORRECTION APPROACHES

Branching ratio (BR). In 2019, Fei utilized the BR method to discern the non-self-absorbed lines in calibration-free laser-induced breakdown spectroscopy (CF-LIBS). The BR parameter is determined by the ratio of the intensities of two lines, denoted as I_1 and I_2, that stem from a common upper energy level. The theoretical BR and measured BR can be expressed as:

$$BR = \frac{A_1}{A_2} \tag{21}$$

$$BR = \frac{I_1/\varepsilon(\lambda_1)}{I_2/\varepsilon(\lambda_2)} \tag{22}$$

Where A is the transition probability; ε is the spectral responsivity. If self-absorption effects are not present in either of the lines, the measured BR will align with the theoretical value. Nevertheless, it is important to recognize that the equivalent value may also be impacted by the degree of self-absorption in a proportional manner. The author suggests that adjusting experimental parameters can resolve this problem, as variations in the plasma state can affect particle density and distribution. Thus, the degree of self-absorption may fluctuate if these lines are susceptible to self-absorbed.

Following the application of this method for selecting CF-LIBS lines, it was observed that the error for the primary element Ni in the nickel-base alloy varied between 0.04% and 19.03%. T. Fei *et al.*[54] determined that the Ti I and Ti II spectral lines of LIBS could be corrected using BR by optimizing the laser energy and delay time. In 2021, Medina *et al.* [55] utilized the BR approach to evaluate optically thin conditions in LIBS, in conjunction with the principles of the Self-Absorption-Free LIBS (SAF-LIBS) technique.

It is possible to detect the presence or absence of self-absorption in either or both of the lines, despite a notable deviation in the line intensity ratio from the anticipated value within the optically thin plasma limit. Moreover, upon confirming that one line in the ratio is devoid of self-absorption, a timely assessment can be conducted to ascertain its effect on the other line. This method offers the advantage of being unaffected by plasma temperature, despite the three perquisites in CF-LIBS. However, its shortcoming is blatant as it requires the presence of lines that correspond to the same upper energy level. Therefore, in experimental investigations necessitate numerous spectral lines.

SA parameter for correction (SAC). The principle of the SA coefficient method, which corrects the self-absorption effect, is detailed in the section of "EVALUATION", and only some examples of applications will be presented next. Many LIBS experiments have utilized the SAC method to rectify distorted profiles by determining the line width and evaluating self-absorption for further study. Mansour *et al.*[56] examined the impact of self-absorption on the electron temperature and compensated for the Al lines. They conducted a comparison between the electron number density obtained from the collected spectra and the optically thin H_α line. As previously mentioned, this approach requires the determination of both the electron number density and the Stark coefficient. Hence, Bredice *et al.* [57] presented an alternative approach in cases where the Stark coefficient is not accessible, investigating the potential of the SA parameter defined as the ratio of line intensity. Additionally, the authors proposed a method that solely considers line intensity for determining the SA coefficient parameter.[44]

In addition, another approach to determining the SA coefficient parameter was presented by Pace *et al.*[58, 59], employed a radiation transfer-based calculation of optical thickness. They have been utilized this method to correct the Mg spectrum abstained from a calcium hydroxide sample, under different laser energy conditions. There is another relative SA coefficient method was proposed by Y. Zhang *et al.* [60] to improve the accuracy of quantification in underwater LIBS analysis.

Duplicating mirror. Duplicating mirror method can not only check whether it produces a self-absorption effect but also establish a model to correct self-absorption. In 2014, Burger *et al.* [61] combined a CCD camera with a mirror behind it and proposed a relatively simple expression for self-absorption correction. According to this expression, the corrected self-absorption-free spectrum F_0 can be calculated by Eq. (23):

$$F_0 = \frac{2F_1}{1+\frac{F_2-F_1}{GF_1}} \tag{23}$$

Where the subscripts 1 and 2 correspond to the spectral intensity without mirror and the spectral intensity with mirror, respectively; G is the spectral intensity loss factor, which considers factors such as specular reflectivity, lens steric angle, and transmittance.

When assessing the self-absorption effect through the duplicating mirror technique, spherical mirrors are employed, while planar mirrors are utilized for correcting the self-absorption effect. [61-63] Burger *et al.* [61] applied the duplicating mirror technique to correct emission lines before performing the inverse Abel transform. However, it should be noted that the approach is suitable for scenarios with weak self-absorption due to the approximation

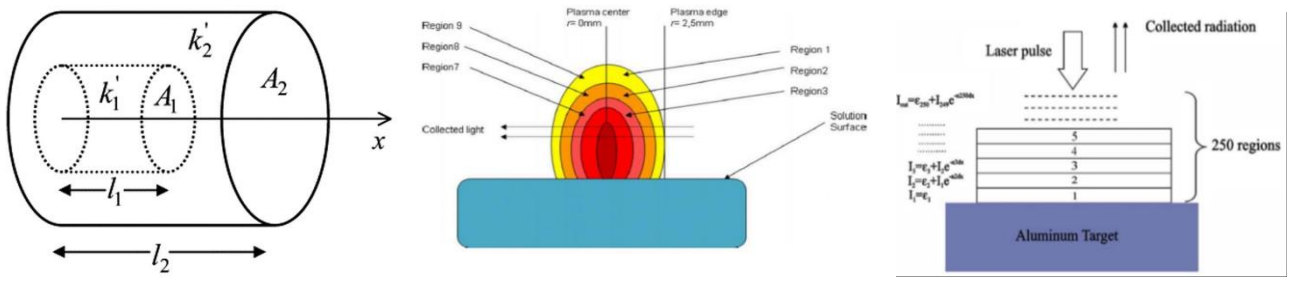

Fig. 5 Inhomogeneous plasma models: (a) a columnar model with two layers. Reprinted from[154]; (b) model with 5 layers. Reproduced from[74]; (c) model with 250 layers.

employed. Cvejic *et al.* [62] also utilized it to correct self-absorbed Mg spectra to obtain the Stark broadening.

Escape factor. The concept of the escape factor was first proposed by Darwin *et al.* [64] and then further explored by Irons [65-68]. Differing from SA coefficient, the escape factor is defined as the ratio of self-absorbed line intensity that escapes from the optically thick plasma compared to the emission of the same line under optically thin conditions. Given that the spectral profile conforms to a Lorentzian distribution, the correlation between the escape factor and SA coefficient parameter can be expressed as follows:

$$\Lambda_r = \int_0^{\infty} P(\lambda) \exp\left[-\tau_0 \frac{P(\lambda)}{P_0(\lambda)}\right] d\lambda = (SA)^{\beta} \qquad (24)$$

Where, $P(\lambda)$ is the line profile after normalization; $\beta = 0.46$ as usual; τ_0 stands for the monochromatic optical depth.

Numerous research teams have utilized this method for quantitative analysis. For example, in 2008, Hannah *et al.* [69] utilized the escape factor to quantify the Na concentration in the Mediterranean Sea and the Oceans. They also applied this method to evaluate the self-absorption effect in the Ca I 422.67 nm and Ca II lines (393.37 nm, 396.85 nm) in $CaCl_2$ solutions[70]. Their finding indicated that the self-absorption effect becomes negligible when the sodium concentration is less than 10^{-3}% and the plasma temperature exceeds 10000 K.

Additionally, significant research has been carried out by Parigger *et al.*[63] , who combined several methods, including the duplicating mirror and escape factor method, to examine the influence of atmospheric pressure gas and highly pure hydrogen gas on plasma characteristics. It should be note that this approach also necessitates the assessment of electron number density, plasma temperature and line broadening parameters. The study conducted by J. He and L. Xia [71] offers an in-depth discussion on of the escape factor concept, along with the computational methods for determining specific escape factors.

Inhomogeneous plasma models. Various plasma models have been created to elucidate the mechanism of self-absorption. One of the simplest models, Inhomogeneous plasma models, was proposed by Lazic *et al.* [72] for analyzing soil and sediment measurements. In their model, the plasma is assumed as a rod of length L, divided into N equally sized layers of δL length. The authors provide a mathematical formula to correlate the measured species concentration C_m and the actual value C_E, as follows:

$$C_m = \frac{a_1}{a_2}[1 - \exp(-a_2 C_E)] + a_3 C_E \qquad (25)$$

In experiments, the coefficient a_i $(i = 1,2,3)$ can be determined. The first term in the Eq. (25) represents the optically thick plasma, while the second term accounts for the optically thin plasma. The authors noted that, the measurement errors for most elements in the analyte were dropped by less than 20%. For Na, Al, and Zn, however, the errors were closer to 40%. Hansen *et al.* [73] also applied a plasma model with two homogeneous zones along the line of sight through the plasma center to analyze the LIBS spectrum obtained from the Martian surface.

Figure 5 displays various inhomogeneous plasma models characterized by a shared feature. The intensity of the emission spectral intensity in the inner layers plus the intensity of absorption spectrum in the periphery is the total spectral intensity. In line with the fundamental principles of inhomogeneous plasma models, the plasma plume is conceptualized as a series of distinct layers, each of which is divided into numerous very thin layers for analysis, so that the individual layers can be considered as optically thin.

As an illustration, in the homogeneous scenario depicted in Fig. 5(b), the plasma plume was segmented into five concentric spherical layers[74]. Therefore, the intensity of the collected spectrum is the sum of the spectral intensities of these five layers, each with an opacity of 0.5 mm and was calculated by:

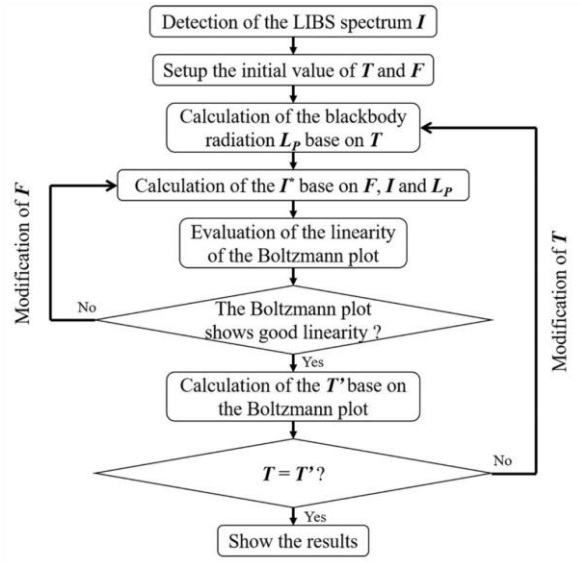

Fig. 6 The flow chart of the BRR-SAC method.

$$I_{total} = \sum_i \left[\frac{I_i}{\tau_i}(1 - e^{-\tau_i})e^{\sum_{j>i} - \tau_i}\right] \quad (26)$$

Where the indexes i and j are represent specific regions. The layers in these models can be divided into smaller, more continuous segments, enabling the calculation of line intensity during integration. However, this method assumes a consistent distribution of certain properties within the plasma plume without significant variations.

Internal reference for Self-Absorption Correction (IRSAC). In 2009, Sun *et al.* [75] utilized the IRSAC method to correct the self-absorption effect in CF-LIBS. They identified a spectral line for each element with a high excitation energy level to minimize self-absorption effects. By comparing this selected reference line with other lines from the same element, they were able to assess and rectify the degree of self-absorption using a regression algorithm. The authors observed an improvement in the linearity between line intensity and concentration following the implementation of this correction method. This method has been widely adopted by other researchers,[76, 77] as evidenced by its successful application in the research conducted by Ramezanian.[78] Their work not only involved estimating the hardness of an alloy composed of Fe, Cr and Ni but also resulted in a notable improvement in the correlation coefficient (up to 90%) between the intensity ratio of the Cr ionic line to the atomic line and the hardness. Moreover, Z. Farooq *et al* [79] employed the IRSAC method to calibrate the profiles of self-absorption effects on the spectral lines of Ni and Sn nanoparticles in LIBS spectra.

In various studies, authors have utilized the IRSAC method along with other beneficial tools to improve the efficiency of CF-LIBS. For example, in a study on coal analysis conducted by S. M. Z. Iqbal *et al.* [80], CF-LIBS, CF-LIBS with IRSAC, and CF-LIBS with density correction were employed to solve the self-absorption effect in CF-LIBS. These methodologies were evaluated against the certified composition, with the outcomes indicating that CF-LIBS with IRSAC yielded the most favorable results. Farooq *et al.*[81] utilized the IRSAC method in conjunction with an electron number density conservation model to analyze laser ablation and the evolution of plasma properties. Additionally, Borges *et al.*[82] employed the line width method to evaluate self-absorption of a particular analyzed line and opted for the IRSAC method for lines comparable energy levels.

The limitations of the IRSAC approach should be noticed. It can be challenging to identify optically thin line for each element; as reference lines are typically weak. And the selection procedure seriously relies on experiential knowledge. To improve the accuracy of the IRSAC method, Dong *et al.* [83] introduced the internal reference-external standard with iteration correction (IRESIC) technology. This method involves using a standard sample for comparison with unknown targets and employs the genetic algorithm as a supplementary tool.

Blackbody reference radiation for Self-Absorption Correction (BRR-SAC). In a recent technique proposed by Li *et al.* [84] in 2019, the detected spectrum was analyzed by comparing it to the theoretical blackbody emission. Then calculate the plasma temperature *T* and the collection efficiency *F* of the optical system respectively, and continuously assign and modify the two parameters, and finally realize self-absorption correction. *F* is mainly related to the diameter of optical fiber, numerical aperture of optical fiber, lens size and other factors. The spectrometer and optical fiber used in the experiment are calibrated, so the collection efficiency of the optical system is mainly affected by the experimental optical collection system, and it is generally difficult to determine the value of *F*. Plasma temperature *T* is related to the experimental environment, laser energy and other factors, and can be calculated in combination with the Boltzmann plot, so the impact of plasma is relatively small.

According to the plasma temperature *T* and the optical system collection efficiency *F*, calculate the black body radiation intensity $I_p(\lambda, F, T)$, the formula is:

$$I_p(\lambda, F, T) = FL_p(\lambda, T) \quad (27)$$

The above formula has only two variables, *F* and *T*, which determine the correction result of spectral intensity of the spectral line. The flow chart of the BRR-SAC method is shown in Fig. 6. It is the Boltzmann diagram that associates with the correction spectral line strength and plasma temperature *T*, so the characteristic spectral line strength of any element in the sample to be tested is selected for Boltzmann fitting. Continuously adjust the plasma temperature *T* and the value of the optical system

Table 1 Comparison of several self-absorption compensation and correction approaches

Approaches	Advantages	Disadvantages
Branching ratio	(i) Simple and quick to tell the line self-absorbed or not	(i) Medium accuracy
SA parameter	(i) Simple for calculation	(i) Requires the knowledge of electron number density and the Stark coefficient.
Duplicating mirror	(i) Simple to detect self-absorption.	(i) Complex for calculation (ii) Consideration for several optical parameters (iii) Susceptible to the kinds of line profile
Escape factor	(i) Evaluates the self-absorption quantitatively.	(i) Requires the knowledge of some plasma parameters
Inhomogeneous plasma models	(i) Improve the stability of the quantitative analysis	(i) Complex for calculation (ii) Incapable for some elements
IRSAC	(i) Simple for calculation	(i) Strictly requires to collect the non-self-absorbed line for every element investigated
BRR-SAC	(i) Simple for calculation (ii) No requirements for the knowledge of broadening parameters	(i) Unremarkable improvement in the linearity of the Boltzmann plot

collection efficiency parameter F. When the linear correlation coefficient R^2 of the Boltzmann diagram is the maximum value, the intensity after self-absorption correction $I^*(\lambda, F, T)$ is obtained:

$$I^*(\lambda, F, T) = FL_p(\lambda)\tau(\lambda) \tag{28}$$

When the optical depth is small enough, $I \approx I^*$. In the context of the CF-LIBS approach, ensuring optically thin plasma is crucial for validity. To address challenges posed by optically thick conditions with high $\tau(\lambda)$ values, the authors extrapolated the optical depth by analyzing the spectrum $I(\lambda)$ and blackbody radiation intensity to prevent inaccuracies.

Finally, there are plenty of similar methods that have been developed based on a reference perspective. For instance, the one-point calibration CF-LIBS (OPC) was proposed by Cavalcanti *et al.*,[85] which involved adjusting the relative efficiency correction factor and theoretical parameters to correct the self-absorption effect and enhance the performance of CF-LIBS in practical settings. Fu *et al.*[83] introduced the standard reference line (SRL) approach. Additionally, they combined the OPC with SRL techniques to analyze six elements present in samples of stainless and heat-resistant steel.[86]

Others. In the research conducted by A. N. Kadachi *et al.*,[87] various self-absorption correction methods (SAC) were introduced to overcome the inherent spectroscopic limitation of plasma emission due to the self-absorption effect of light emitted by the plasma generated from the from the laser beam's interaction with the sample surface. The SAC method involved determining the electron density ratio by analyzing the emission of low-concentration elements in the air atmosphere. These elements exhibit non-absorbing lines due to Stark broadening of absorption lines. The improvement accuracy ofquantitative analysis through SAC, particularly in (LIBS) of single ionization emission lines, was confirmed by validating the process with two additional standards. F. Deng *et al.*[88] proposed an single-sample calibration-LIBS (SSC-LIBS) with SAC method to enhance the quantitative analysis of precious sample. The SSC-LIBS with SAC involves adjusting the intensity ratios of spectral lines in SSC-LIBS calculations using the relative self-absorption coefficient, K, without the need for complex sample preparation details. Experimental findings demonstrate significant improvements in the quantitative accuracy and applicability of SSC-LIBS with SAC over conventional SSC-LIBS, thereby greatly facilitating the widespread application of SSC-LIBS. Furthermore, Hu *et al.*[89] introduced a novel self-absorption correction technique for single-point calibration LIBS (OPC-LIBS). This method aims to overcome the challenge posed by self-absorption effects on measurement accuracy in LIBS. It employs an algorithm to correct for self-absorption without the necessity of computing self-absorption coefficients.

Summary

Different methods for correcting self-absorption effects vary in their applicability and come with their own set of pros and cons. Hance, depending on the specific experimental conditions, appropriate methods can be chosen to effectively correct the self-absorption effect. Table 1 presents a comparative analysis of these approaches, offering a thorough assessment to assist in determining the most appropriate approach for a particular situation.

SUPPRESSION METHODS

Self-absorption in LIBS technology continues to pose a persistent challenge. However, through conducting targeted scholarly research, it is possible to effectively address this issue by making precise modifications to certain experimental parameters or by employing external devices.

Experimental Parameters

Laser parameters. During the laser-material interaction, the laser parameters exert a decisive influence on the formation and evolution of plasma, making them key factors for reducing the LIBS self-absorption effects. Adjusting parameters such as laser wavelength, energy and pulse width can effectively minimize the self-absorption effect.[33] For example, a study conducted by Díaz *et al.*[90] demonstrated that a shorter laser wavelength (355 nm) performed better compared to a longer wavelength (1064 nm) in terms of ablation. These parameters also play a crucial role in the evolution of electron temperature and electron number density[91]. Numerous literature sources have emphasized the correlation between self-absorption and laser wavelength, as exemplified in the recent study by Cui *et al.*[92] Notably, Lee *et al.*[93] have also explored the wavelength-dependent characteristics of self-absorption effect. Their research revealed strong self-absorption effects in Cu I lines when employing a Nd: YAG laser operating at a wavelength of 1064 nm; while a less pronounced self-absorption effect was observed when employing a laser operating at 532 nm. The influence of laser wavelength on self-absorption can be explained through the shock wave-driven excitation mechanism. Generally, shorter wavelengths generate more potent and rapid shock waves, thereby facilitating the excitation of atoms in the peripheral regions with greater efficiency. In the context of underwater LIBS, employing long pulses also results in reduced self-absorption effects compared to short pulses, as meticulously illustrated in the research by Li *et al.*[94]

Sabri *et al.*[95]conducted a study on the self-absorption resonant lines. The authors observed a decrease in the self-absorption effect as laser intensity increased under atmospheric pressure conditions. Another investigation by Kuzuya and Aranami[96] examined the impact of high copper content in metal alloys at atmospheric pressure. They findings revealed that self-absorption could be significantly reduced by enhancing laser intensity in the initial stages.[97, 98] Recently, W. Ke *et al.* [99] found that the self-absorption effect of the spectral line Cu I 521.82 nm was intensified as the delay time and laser energy increased, irrespective of the vacuum conditions being high or low.

Maintaining the laser-related parameters mentioned above, altering the angle between the laser beam and the sample can also attenuate the self-absorption effect. In 2021, Pardede *et al*[100] proposed a new approach involving parallel laser irradiation to suppress self-absorption in LIBS. This technique creates a transien rarefied atmosphere near the sample's surface, facilitating the generation of homogenous sample plasma.

Background gags conditions. Given the complex evolution of self-absorption in varying background gas conditions, it is pertinent to commence by examining a selection of typical works. Lee *et al.* [93] investigated the influence of different surrounding gases (argon, neon and helium) on copper atomic lines at wavelengths of 510.55 nm, 515.32 nm and 521.82 nm. The research revealed a significant alteration in the spectral profile when exposed to helium ambient gas, with the least self-absorption observed in the presence of argon background gas.[101] Rinda *et al* demonstrated that the self-absorption effect could be ignored by leveraging the He metastable excited state (He *) and employing the Penning-like energy transfer mechanism for delayed excitation of the ablated analyte atoms [102]. Their findings suggest that the attenuation of self-absorption manifests as apparent when helium is used as the ambient gas. Wang *et al*[103] simulated and verified the radiation characteristics and self-absorption effect of laser-induced plasmas under different background gases. They found that the self-absorption effect can be alleviated by utilizing a gas with a low ionization potential.

External pressure. In analyzing the correlation between the background gas pressure and self-absorption, it is essential to consider the impact of ambient gas pressure on the plasma plume. Generally speaking, lower-pressure environments typically decreased self-absorption, while higher-pressure environments magnify self-absorption effect.

For instance, Suliyanti *et al.* [104] conducted a study on plasma emission induced from a solid organic analyte at low-pressure. They findings revealed that emission lines generated at lower gas pressure displayed more distinct profiles compared to those generated at higher-pressure This observation was attributed to the diminished ablation process that occurs at lower pressure. Additionally, the utilization of laser shielding,[105, 106] which reduces the ablation rate, resulting in self-absorption attenuation. Notably, Hao *et al.* [107] investigated the impact of background gas pressure on resonance lines (Mn I 403.1 nm and Cu I 327.4 nm). Their findings manifested that the effectiveness of self-absorption suppression was enhanced when the gas pressure was kept below 20 kPa. Intense laser shielding raises the temperature of plasma. Consequently, this reduces the presence of cold particles within the plasma. Zhang et al. [108] conducted a study on the impact of pressure on the self-absorption effect in underwater LIBS. The research covered a range of 0.1 MPa to 40 MPa, with outcomes aligning with previous findings, indicating that reduced pressures were associated with diminished self-absorption effect. It is worth to note that, among the extensive research conducted on the effects of ambient atmosphere and high pressure on spectral properties, Yuan *et al.* [109]investigated the influence of low air pressure on the characteristics of long pulse width (hundred nanosecond level) LIBS. Their research revealed two noteworthy findings. Firstly, a reduction in atmospheric pressure led to a notable decrease in the self-absorption of the matrix element copper. Additionally, the signal-to-background ratio of spectral lines containing low concentrations of iron increases, plasma temperature rose, and spectral lines broadening decreased.[110] In line with most instances, Ning *et al.*[111] demonstrated a significant decrease in self-

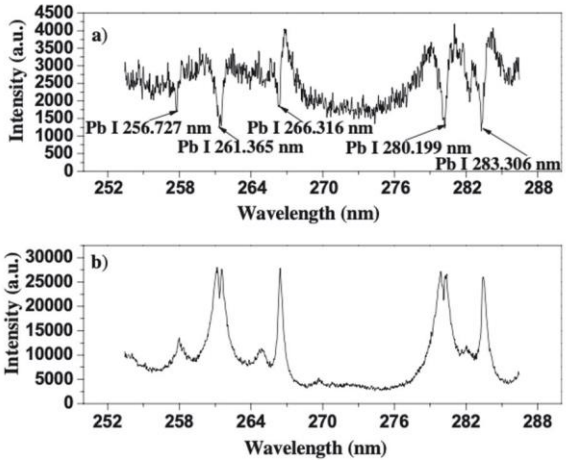

Fig. 7 Self-absorption recorded at the delay time of 200 ns in the range of 255 nm to 285 nm. The positions are 6.9 cm (a) and 6.1 cm (b).

absorption of copper alloy plasma by reducing ambient pressure at laser energy between 0 and 100 mJ. LIBS is also commonly employed in soil analysis, with the manipulation of the soil pressure at low levels proving effectively suppressed self-absorption effects on the results[108]. Additional valuable research can be referenced to explore self-absorption effect in high air pressure conditions. Similarly, an analysis from the perspectives of plasma expansion and confinement resulting from shielding offers valuable insights. Furthermore, an intriguing experiment was carried out in the context of in the Martian atmosphere case.[112]

Temporal and spatial evolution. The level of self-absorption is also influenced by the delay time and location of spectra acquisition. Diao *et al.* [113] conducted a study investigating the impact of the distance between the focal length of the lens and the sample on self-absorption. They observed that self-absorption effects were stronger when the target was positioned farther from the focus point as opposed to being placed in front of it. Fig. 7 depicts the findings of resonant lines between 255 nm and 285 nm in the Pb atom, obtained at a consistent delay time of 200 ns but at different positions (6.9 cm and 6.1 cm). The authors have attributed these phenomena to the presence of breakdown air between the lens and target. When the sample is positioned below the focus, the laser initiates the breakdown of air, resulting in the formation of a gas plasma above the sample. Due to the absorption characteristics of the gas plasma, a reduction in laser energy reaching the sample surface. Consequently, a higher proportion of neutral atoms contribute to self-absorption effects.

Fu *et al.*[114] studied the influence of the distance between the lens and the sample on the self-absorption of the Pb atom line at 368.3 nm. Their findings indicated that positioning the target 6 mm back the focus resulted in self-reversal effect at delay times below 300 ns. Conversely, a minimal self-absorption effect was observed in the line profile when the sample was placed at the focus point. In a distinct study, Pardede *et al.*[97] employed a parallel-emitting nanosecond Nd: YAG laser operating at 1064 nm, which was accurately focused near the sample surface at a distance of 1 mm. In atmospheric pressure air condition, the self-absorption effect observed in the acquired resonance spectra was determined to be minimal. Remarkably, Marpaung *et al.* [115] introduced a method called defocused laser irradiation to vanish the self-reversal in the line of K I 766.4 nm, K I 769.9 nm, Na I 588.9 nm, and Na I 589.5 nm. Their study involved laser energy, ranging from 10 to 50 mJ, onto the specimen's surface using a flat convex lens with a focal length of 150 mm. By deliberately defocusing the laser beam by approximately -6 mm, they effectively reduced the impact of self-absorption effects in the spectral analysis[115]. Furthermore, Yi *et al.* [116] studied the spatial correlation of self-absorption through spatially resolved LIBS. They collected spectra analysis along the Z-axis at different positions and observed that the a less impacted spectrum by self-absorption could be obtained at lower plasma positions. This phenomenon was not limited to the specific matrix under examination but was also useful in other materials.

The influence of delay time should be considered as another significant factors. As the plasma radiates from the its core outward, the plasma heat and the population of low-energy particles diminish, leading to a relatively low level of self-absorption effect in the initial stage.[117] Additionally, the decrease in energy loss resulting from inelastic collisions with air particles is reduced. Meanwhile, the spectral lines experience ongoing modifications influenced by variables like electron density and plasma temperature, underscoring the importance of capturing them at the precise moment. Tang *et al.*[53] observed a rise in self-absorption over delay time through the ICCD camera. We identified the minimum value of self-absorption coefficient at 200 ns for the Cu I 327.40 nm. Hou *et al.*[118] further examined the lines of Al I 394.4 nm and 396.1 nm noting that these lines remained very narrow until a delay time of 400 ns, after which they regained their original thickness.

Ahmed *et al.*[119] focused on the self-absorption effects of Ca ion lines and determined that in the early stages of plasma evolution, there were almost no spectral lines with self-absorption effects. The most plausible explanation for the phenomenon is attributed to the increased population of low-energy states in the plasma as it cools gradually, which leads to the reabsorption of radiation over time. This process is characterized by an exponential way in both plasma temperature and electron number density. The evolution of these two parameters over time, obtained by averaging the values associated with three resonant lines, is presented in Fig. 8. He *et al.* [120] performed an analysis on the self-absorption effect of Ca, Mg, Al lines in soil. They reported a significant decrease in self-absorption effect within the delay time range of 0.2-1.2, which was linked to the diminishing impact of Stark broadening. Subsequently, a steady state was observed within the delay time range of 1.2-1.6.

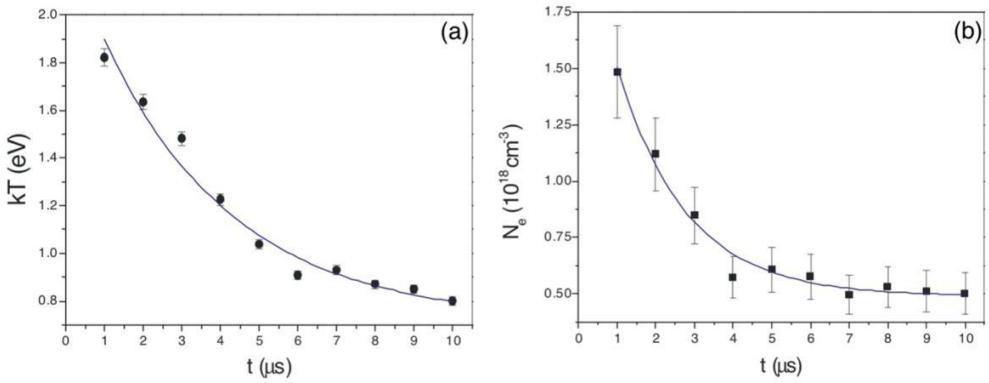

Fig. 8 A diagram of temporal evolution of plasma temperature (a) and electron number density (b).

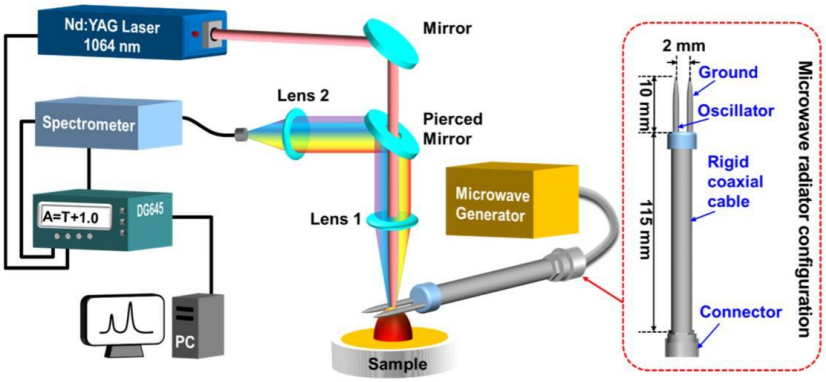

Fig. 9 The schematic experimental set-up of the Microwave-Assisted Excitation LIBS.

It should be pointed out that the results have referred to resonance lines and transitions ending up at low energy levels so far. As for the case of transition featuring high energy levels, things are not that similar.

Selection of spectral liens. Spectral lines can be classified resonant and non-resonant. Typically, resonance spectral lines are more susceptible to self-absorption effects than non-resonant lines. For instance, Chan *et al*[121] found that employing non-resonant lines in LIBS has demonstrated a decrease in self-absorption effects when analyzing gaseous uranium hexafluoride (UF6), in contrast to the method of optimizing experimental parameters and applying self-absorption corrections. However, a research conducted by M. Hornackova *et al*[122] in 2019 utilized non-resonant spectral lines in developing calibration curves to successfully mitigate the impact of self-absorption. They also found the atomic and ion spectral lines observed in identical experimental conditions demonstrate differing levels of self-absorption effects. Aguilera *et al* [123] found that the self-absorption of non-resonant lines with increasing delay time, and this tendency does not diminish until a specific value. However, it is noteworthy that a simulation method proposed by John *et al*[124] revealed that resonant lines exhibit stronger self-absorption compared to non-resonant lines.

External Devices

In addition to optimizing the experimental parameters, mitigating the self-absorption effect can be achieved by the integration or substitution of equipment. Key methods for achieving this include utilizing microwave-assisted excitation, fiber laser-assisted techniques, and energy-state selective resonance excitation. Furthermore, there exist less common external devices that can also assist in reducing the self-absorption effect. A concise description of these auxiliary devices is provided below.

Microwave-Assisted Excitation LIBS (MAE-LIBS). Tang *et al.*[6] designed an experimental setup employing a MAE system to suppress the self-absorption effect in LIBS, as illustrated in Fig. 9. The methodology involves energizing out-layer electrons with microwaves, causing them to an excited state through successive collisions. The effectiveness of this approach contingent upon parameters such as species concentration and upper energy level.

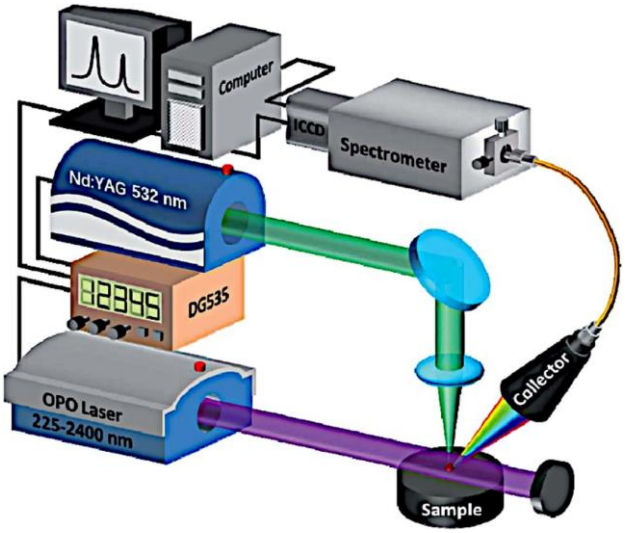

Fig. 10 The schematic experimental set-up of the Laser Stimulated Absorption Excitation LIBS.

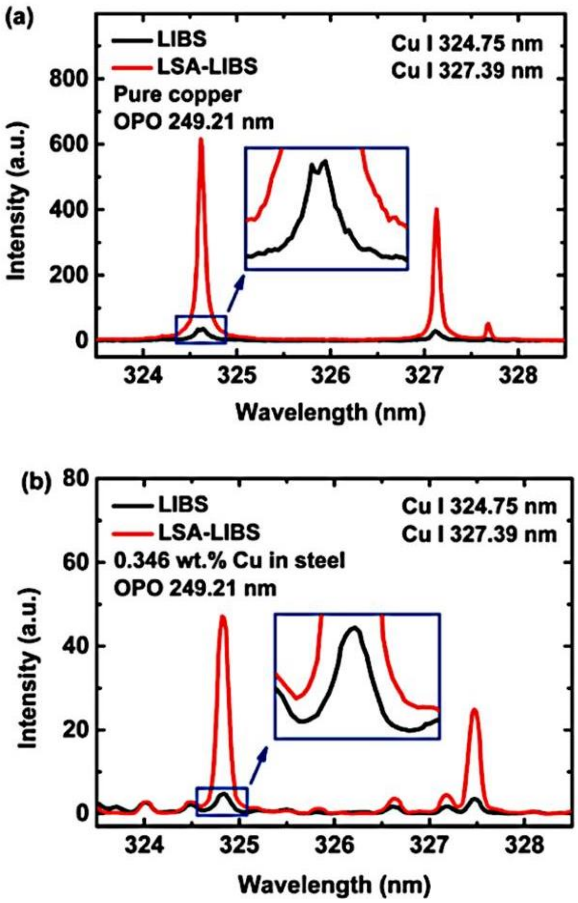

Fig. 11 LIBS (black) and LSA-LIBS (red) spectra were collected from a pure copper sample (a) and a sample featuring 0.346 wt% copper (b).

They utilized MAE-LIBS to alleviate the self-absorbed effects of spectral lines in an analysis of potassium feldspar. The results demonstrated noteworthy reductions in FWHM values for the spectral lines of Na I 589.0 nm, Na I 589.6 nm, K I 766.5 nm and K I 769.9 nm, showing decreases of 43%, 43%, 53%, and 47%, respectively. In addition, Viljanen *et al.* [125] previously employed a wave excitation setup in LIBS, aiming to enhance the plasma emission.

In essence, this strategy involves enhancing the homogeneity of plasma through utilizing collision-induced processes. It is worth noting that there is no requirement to choose specific wavelengths for different elements, indicating the feasibility of multi-elemental analysis. Although this approach presents advantages such as small size, cost-effectiveness, and ease of maintenance, its stability has not been guaranteed. Fortunately, Fan *et al.*[126] proposed a novel single-probe microwave radiator that showcasing a unique heat dissipation design. This new radiator design enhances the accuracy of MA-LIBS devices and mitigates instability concerns stemming from the microwave source. It improves superior microwave power injection and eliminates the self-reversal phenomenon.

Laser Stimulante Absorption LIBS (LSA-LIBS). In 2015, Li *et al.*[127, 128] introduced LSA-LIBS and elucidated that the absence of self-reversal in the spectrum was attributed to the ground-state atoms at the periphery absorb these photons. The experimental setup for LSA-LIBS is illustrated in Fig. 10. Furthermore, a comparative analysis between LSA-LIBS and traditional LIBS was conducted to assess the quantification of minor elements concentration, as presented in Fig. 11.

Subsequently, a new advancement was achieved by integrating spatial resolution into the framework, leading to the creation of Space-Resolved Laser Saturation Absorption (SS-LSA)-LIBS[129]. By employing an Optical Parametric Oscillator (OPO) laser pulse, the plasma in a micro-alloyed steel specimen was excited at various positions. Subsequently, calibration curves were generated for Cu I 324.75 nm and Cr I 425.43 nm at varying excitation points. The results indicated that this approach performed better in enhancing fluorescence intensity compared LSA-LIBS. In addition, the authors performed a comparative evaluation of SS-LSA-LIBS, LIBS and LSA-LIBS regarding their quantitative analysis capabilities. This evaluation included assessing accuracy through root mean square error cross validation (RMSECV), examining the self-absorption effects with the self-absorption factor α, and determining linearity using R^2. The Quantitative results presented in Table 2 led to the conclusion that SS-LSA-LIBS effectively mitigates self-absorption and improves accuracy to a higher degree than other methods.

The increase signal intensity can be attributed to various factors, including the reduction of the self-absorption effect and the greater population of the upper energy level. This is facilitated by redistributing the population of the upper energy level to adjacent

Table 2. Comparison of different techniques (LIBS, LSA-LIBS and SS-LSA-LIBS) in the effectiveness of analysis, in terms of the RMSECV, the linear parameter and the self-absorption factor. Reproduced from [53]

Approches	RMSECV (wt. %)		Linear			
	Cu	Cr	Cu	Cr	Cu	Cr
LIBS	0.042	0.356	0.772	0.763	10.701	1.598
LSA-LIBS	0.020	0.051	0.972	0.997	3.308	0.130
SS-LSA-LIBS	0.008	0.047	0.993	0.999	1.527	0.116

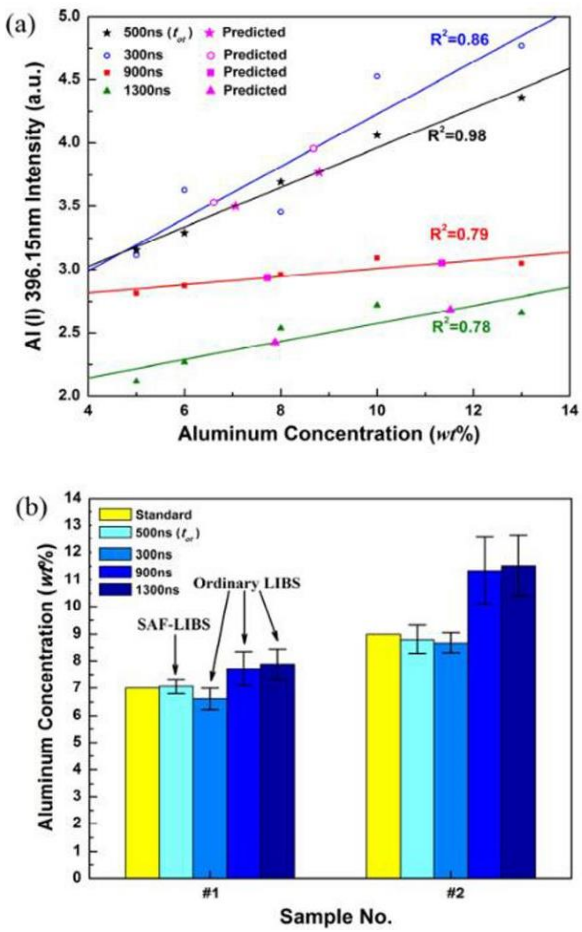

Fig. 12 (a) Calibration curves of Al; (b) comparison between SAF-LIBS and LIBS.

levels through collisional interactions. Additionally, Tang *et al.*[130] successfully eliminated spectral interference and self-absorption issues through the application of LSA-LIBS. The researchers documented a substantial reduction 85% in the self-absorption factor for Ni I 341.47 nm, along with a remarkable 83% decrease in the average relative error. Chen *et al*[131] employed target-enhanced orthogonal double-pulse LIBS in combination with a modified one-point calibration method to address the challenge of self-absorption when analyzing bismuth brass with conventional CF-LIBS

Fiber Laser Ablation LIBS (FLA-LIBS). In 2019, Xiong *et al.* [132] introduced the FLA-LIBS method, employing a fiber laser to suppress self-absorption in LIBS experiments. They applied this technique to perform spectroscopic analysis on a chromite sample. By ablating a portion of the sample into gas, subsequently exciting it with a Nd: YAG laser. Authors also assessed the self-absorption by measuring the SA coefficient parameter. Interestingly, the SA coefficient for Mg and Ca in the sample was found to be significantly closer to 1 compared to traditional LIBS. By applying the FLA-LIBS, the linearity of the calibration curves for Mg and Ca were enhanced significantly from 0.841 and 0.847 to 0.954 and 0.895, respectively. The improvement can be attributed to the presence of aerosol state, which aid in maintaining LTE within the plasma. Consequently, atoms in their fundamental state can more effectively absorb energy, leading to a transition towards a thinner and more optimal state. FLA-LIBS offers a cost-effective way to suppressing self-absorption. But further investigation is necessary to explore the underlying mechanism of this method in greater depth.

SAF-LIBS. Hou *et al.* [118] introduced a method aimed to accurately measure the intensity of the quasi-optically thin state, by comparing the intensity ratio of spectral lines in a doublet with theoretical predictions. This method allowed for the determination of the optimal delay time and integration time. By analyzing the intensity ratio of spectral lines from transitions with closely matched upper energy levels at different delay times, the authors were able to identify the time window corresponding to an equivalent optically thin condition:

$$C_{n\Delta t,meas} = \frac{I^1_{n\Delta t,meas}}{I^2_{n\Delta t,meas}}, (n = 1,2,3 \dots) \qquad (29)$$

Where Δt is a fixed time interval; indexes 1 and 2 correspond to the line intensities of two investigated lines. Theoretical value of the ratio can be derived from:

$$\frac{I_{1,0}}{I_{2,0}} = \left(\frac{\lambda_{nm,Z}}{\lambda_{ki,Z}}\right)\left(\frac{A_{ki,Z}}{A_{nm,Z}}\right)\left(\frac{g_{k,Z}}{g_{n,Z}}\right)\exp\left(\frac{E_{n,Z}-E_{k,Z}}{K_B T}\right) \qquad (30)$$

Where $I_{1,0}$, $I_{2,0}$ represent the line intensities corresponding to the $n \rightarrow m$ and $k \rightarrow i$ transitions, respectively; λ, A, Z, g E, K_B and T are the ionized state, transition wavelength, transition probability, degeneracy, energy level, Boltzmann and plasma temperature.

When the aforementioned lines 1, 2 originated from the specified transitions, the exponential function on the right-hand

side of the equation approached a value of one. Consequently, the time window can be determined as follows:

$$n_{win} = INT\Delta t \tag{31}$$

The authors conducted a comparative analysis between SAF-LIBS and conventional LIBS to evaluate the accuracy of elemental concentration analysis. The results of this comparison are shown in Fig. 12. However, it is noted that the general SAF-LIBS technique has a limited range for elemental concentration analysis. To solve this limitation, an enhanced method was devised by integrating resonant and non-resonant lines. As anticipated, the new SAF-LIBS technique has shown improved measurement capabilities across a broader spectrum of elemental concentrations while maintaining optimal sensitivity (limit of detection: LOD = 1.35 ppm). It is important to note that this technique necessitates a relatively high spectral resolution due to the incorporation of double lines. Moreover, t uncertainty in transition probability may impact the determination of the optimal delay time.

In 2019, Hou *et al*. [133] utilized SAF-LIBS to determine the optimal acquisition delay time by analyzing the bilinear intensity ratio of the analysis unit at a specific transition wavelength. This approach facilitates the direct acquisition of the appropriate optical thin lines, thereby mitigating the impact of self-absorption on quantitative measurements. To further explore the effect of self-absorption, researchers utilized a SAF-LIBS technique that integrates resonant and non-resonant doublets.[134] To enhance practicality, reduce costs, and scale down the SAF-LIBS system, Li *et al*. [135] proposed a new quantitative method called high repetition rate acoustic grating controlled SAF-LIBS. This method incorporates an acousto-optic (AOM) device as a grating pulse converter and utilizes a high repetition rate diode-pumped solid-state laser for excitation. Consequently, the utilization of miniaturized CCD spectrometers becomes viable, resulting in a reduction system cost and dimensions, enhancing its utility practical settings.

Double pulse LIBS (DP-LIBS). In the beginning, DP-LIBS was used to enhance the spectral signal intensity. Nevertheless, subsequent research has revealed its potential for mitigating the self-absorption effect. In 2020, Karnadi *et al*. [136] investigated the correction of self-absorption effects using orthogonal DP-LIBS in an atmospheric setting In contrast to FLA-LIBS, Karnadi *et al*. [136] used two Nd-YAG-pulsed laser. The first laser pulse was directed parallel to and focused above the sample surface, then the second pulse was aimed perpendicularly at the sample surface for ablation. The results show that this technique effectively eliminates the spectral line auto-etching of K in the sample within the concentration range of 0.82-26.2 wt%, achieving an SA coefficient near unity. Guillermo Quintana-Silva *et al*. [137] applied orthogonal reheat DP-LIBS to analyze the elemental composition of CdTe thin-film samples, which shows that compared with SP-LIBS, a significantly enhanced in emission intensity of DP-LIBS and the self-absorption effect is significantly reduced. In addition, Wang *et al*.[138] used DP-LIBS to study the self-absorption effect. Through varying pulse energies and delay times, they deduced that the self-absorption of the spectral signals generated by the second laser beam can be suppressed by the presence of the first laser beam and the self-absorption effect is closely related to the delay time while keeping the energy of the first laser beam constant. Chen *et al*.[131] used target-enhanced DP-LIBS in combination with a modified single-point calibration method to effectively mitigate self-absorption effects and attain accurate quantitative elemental analysis of bismuth brass with a relative error of under 4%.

Others. Furthermore, aside from the aforementioned four common auxiliary instruments, there are additional techniques such as laser induced fluorescence (LIBS-LIF) combined with standard additive method (SAM) to assist LIBS,[139] aluminum femtosecond LIBS with spark discharge, fiber laser ablation LIBS (FLA-LIBS),[132] and optically thin LIBS (OT-LIBS).[140]

Summary

There are three subsections in this section aimed at mitigating the impact of self-absorption effects at its core. The first subsection concentrates on optimizing experimental parameter, offering a cost-effective and easily implementable approach. The subsequent part selection of spectral lines to reduce the impact of self-absorption is examined. Lastly, delves into leveraging external equipment assistance. Table 2 provides a summary of the advantages and disadvantages linked to these methods.

EXPLOITATION

Self-absorption has traditionally posed a challenge in the quantitative analysis of LIBS, since its disruption interference with the linear correlation between line intensity and elemental concentration. However, a new perspective has been adopted by scholars, utilizing the inherent information present in spectral lines exhibiting self-absorption effects. By deriving valuable plasma characteristic parameters, such as temperature, electron number density, and Stark broadening, from these lines, and successfully improved the quantitative analysis capabilities of LIBS. This transformative strategy has successfully transformed the substantial drawback of self-absorption in LIBS into a favorable advantage that can be leveraged to enhance analytical capabilities. Many studies have focused on utilizing self-absorption to analyze plasma characteristics and establish spectroscopic parameters.[141] For example, the SA coefficient method,[142] as previously outlined, allows for the determination of the Stark broadening parameter.

Columnar density method. In 2012, Cristoforetti and Tognoni[143] introduced the columnar density method, which can determining the plasma temperature and the absolute number densities of plasma species. This method involves utilizing a modified version of the Saha-Eggert equation to describe the equilibrium distribution of different ionization stages:

$$\frac{n_i^{II}}{g_i^{II}} = \frac{2(2\pi m k_B T)^{\frac{3}{2}}}{n_e h^3}\frac{n^I}{Z^I}\exp\left[-\frac{E_i^{II}+E_{ion}-\Delta E_{ion}}{k_B T}\right] \quad (32)$$

Where the indexes I and II represent the neutral and ionized species, Z is the partition function, E_i^{II} is the energy of lower level of the ionic transition, E_{ion} stands for the first ionization energy, and ΔE_{ion} represents the dropping of the ionization energy, which is typically insignificant compared to the sum of $E_i^{II} + E_{ion}$ in most cases LIBS. Therefore, when Eq. (32) is multiplied by the variable l, it is prepared for representation on the Saha-Boltzmann plot in the usual manner:

$$y = -\frac{1}{k_B T}x + ln\left[\frac{n^I l}{Z^I(T)}\right] \quad (33a)$$

$$x = \begin{cases} E_i^I, & \text{for neutral lines} \\ E_i^{II} + E_{ion}, & \text{for ionic lines} \end{cases} \quad (33b)$$

$$y = \begin{cases} \ln\left(\frac{n_i^I l}{g_i^I}\right), & \text{for neutral lines} \\ \ln\left(\frac{n_i^{II} l}{g_i^{II}}\right) - \ln\left(\frac{2(2\pi m k_B T)^{\frac{3}{2}}}{n_e h^3}\right), & \text{for ionic lines} \end{cases} \quad (33c)$$

Thus, the slope of the line plotted on the Saha-Boltzmann can determine plasma temperature. The new technique offers two notable advantages compared to conventional methods: (i) the abscissa represents lower-level energy values instead of the high-level counterparts; (ii) the accuracy of temperature determination is not compromised by uncertainties stemming from self-absorption effects or the use of an imprecise spectral efficiency curve. However, to ensure accuracy, the authors have introduced the SA coefficient, which is defined in the same manner as Eq. (1). This parameter is crucial for ensuring the method's accuracy, retains accuracy only when $SA \ll 1$, while its effectiveness may be limited if SA parameter closes to 1.

In the research by Safi *et al.*[144], an analysis was conducted to compare the plasma temperature calculated using the conventional Saha-Boltzmann plot and columnar density method,. The findings revealed a significant resemblance in the results obtained from both approaches.

C-Sigma method. Aragon *et al.* [145] introduced the COG method, which involves the formalization of a curve using the Saha-Boltzmann equation. Their study focused on optically thick plasma under the LTE approximation, leading to the derivation of a relationship based on the ratio of line intensity and a parameter denoted as $\boldsymbol{C\sigma}$. And this parameter is defined as the multiplication of the element concentration C and the absorption cross-section that is given by:

$$\sigma = \frac{\kappa}{N_\alpha} = \frac{\tau}{10^{-2}CNl} \quad (34)$$

Where N_α stands for the total number density of a species; κ, τ, are the effective absorption and the effective optical thickness, respectively. It is also possible to evaluate the degree of self-absorption by using the absorption cross-section parameter: a higher value means a severer effect of the self-absorption. In the subsequent analysis, the authors used the Saha equation to determine the ratio between the number density ionized and neutral atoms number densities. Thus, Eq. (35) can be divided into two parts corresponding to the ionic and neutral lines. Based on the assumption that atoms and ions are inhomogeneous layers, a generalized COG method is established:

$$\frac{I}{B(\lambda_0)}\frac{1}{\Delta\lambda_0} = f(C\sigma, \Delta\lambda_0) \approx f(C\sigma) \quad (35)$$

Where $B(\lambda_0)$ represents the Plank blackbody distribution. In contrast to the columnar density approach, the C-sigma method exhibits greater flexibility owing to the independence on $\Delta\lambda_0$. So, his method can be employed to determine the data described in Eq. (35). In addition to obtaining parameters such as plasma electron number density, temperature and particle surface density, in the effective range, predictions of spectral line intensities and self-absorption effects of the spectral lines can be made based on these parameters as well as the atomic information of the spectral lines. For instance, J. A. Aguilera and J. Manrique *et al.*[146-148] utilized the C-sigma method to determine the transition probabilities of the spectral lines in order to avoid self-absorption effects when using LIBS. It is noteworthy that J. A. Aguilera *et al.*[149] verified that C-sigma- LIBS improves the precision of quantitative analysis.

Extended C-sigma method. Due to the limitation of the plasma model used in both methods and the complexity of the numerical calculations, they cannot be applied in many practical cases. Finally, an extended C-Sigma method was proposed that combines the advantages of the column density method and the traditional C-Sigma method and reduces the complexity of the numerical calculations.

Inspecting the right term of Eq. (35), it is evident that complex calculations are required when utilizing this technique in practical experiment. To overcome this shortcoming, in 2019, A. Safi *et al.*[150] integrated the C-Sigma method with the previously discussed columnar density mentioned. This advancement was facilitated by the adoption the Lorentzian profile as the underlying assumption:

$$\frac{\int(1-e^{-\tau(\lambda)})d\lambda}{\int\tau(\lambda)d\lambda} = \frac{\Delta\lambda_0}{\overline{\Delta\lambda}} \quad (36)$$

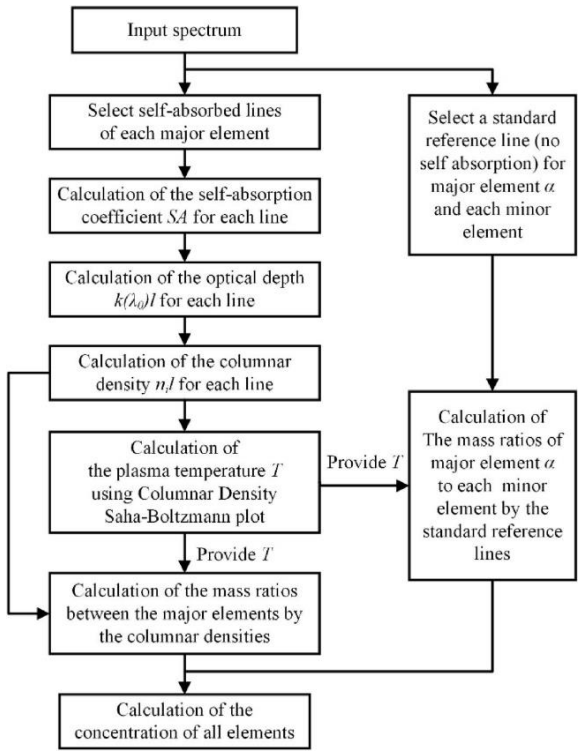

Fig. 13 Flow chart for the calculation of element contents with the use of the CD-SRL technique.

simplified to:

$$\frac{I}{B(\lambda_0)}\frac{\overline{\Delta\lambda}}{(\Delta\lambda_0)^2} = \beta\tau_1 = 10^{-2}\beta CNl\sigma_1 \quad (37)$$

Where, $\overline{\Delta\lambda}$ is the measured line width. The extended C-Sigma can be obtained by plot graphing the left side of Eq. (37) on the y-axis. This approach streamlines the fitting process and enhances the precision of the analysis.

Subsequently, F. Poggialini *et al.*[151] introduced a new time-independent Extended C-sigma approach in LIBS and A. Safi *et al*[152] used it to get the probability of transition and the Stark expansion coefficient of several neutral silver (TIECS) and ionic emission lines.

Self-absorption internal standard (SAIS) model. In accordance with the fundamental principles of the IRSAC approach, it is possible to optimize C-sigma system. He *et al*. [153] proposed a self-absorption internal standard (SAIS) mode, which correlates line intensity with element content, when self-absorption exists. Based on the radiation transfer equation, the line cross section can be expressed by:

$$\sigma_l = k_t r_i \frac{1}{\Delta\lambda_L} \quad (38)$$

Where k_t is equal to Eq. (1), r_i stands for the ionization factor. Then the authors established a connection between the concentration of element denoted as C_α with the integrated intensity of the homogeneous plasma.[154]

$$I = \beta \cdot A \cdot l \cdot L_p \cdot \Delta\lambda_L \cdot \sigma_l \cdot N_\alpha \quad (39a)$$

$$C_\alpha = \frac{N_\alpha m_\alpha}{M} \quad (39b)$$

$$I = \beta \cdot A \cdot l \cdot L_p \cdot \Delta\lambda_L \cdot \sigma_l \cdot \frac{C_\alpha \cdot M}{m_\alpha} \quad (39c)$$

Where M is the sum of the mass of all elements in the sample. L_p is the Planck blackbody distribution. The constant β is influenced by the geometry of the perpendicular radiating surface. When the concentration of a particular element in a sample is already established, as well as other relevant parameters in the Eq. (39c), the element concentration of interest can be by comparing it with the reference content:

$$C_\alpha = C_{\alpha 0} \cdot \frac{I \cdot m_\alpha \cdot L_{p0} \cdot k_{t0} \cdot r_{i0}}{I_0 \cdot m_{\alpha 0} \cdot L_p \cdot k_t \cdot r_i} \quad (40)$$

Where, subscript 0 is corresponding to the element which has been identified prior to the experiment. This method enables the analysis of multi-element concentrations in a nested way. The isotopic abundance of an element affects the center wavelength of the absorption peak of its self-absorption spectrum, so after determining the abundance of an element, the model can be used to quickly identify the elemental species.

Combination of columnar density and standard reference line (CD-SRL). As a mixture methodology, the whole treatment could be separated into two process lines. According to what Z. Hu *et al.*[155] described, the initial process is calculate the mass ratios for major elements present in the sample and the determination of plasma temperature using columnar density method. The subsequent process entails determining the mass ratios of a major element in relation to each minor element, with an optically thin line as the reference line. A detailed procedural outline is depicted in Fig. 13. Apparently, this work exploits the self-absorption effect to determine the plasma temperature as well.

In a study conducted by Z. Hu *et al*[155], researchers demonstrated the effectiveness of utilizing CD-SRL to reduce average errors in CF-LIBS analysis of four aluminum-bronze alloy samples. The application of CF-LIBS with CD-SRL resulted in a decrease in average errors (AEs) from initial values of 3.20%, 3/22%, 3.15% and 3.01% to 0.95%, 1.00%, 1.16% and 1.78%, respectively. It worth noting that while this approach enhances accuracy in the analysis of major elements, there is a potential for compromising the accuracy of minor element concentrations due to the introduction of minor errors during major element analysis.

CONCLUSION

A review of recent advancements in the study of self-absorption effects in LIBS is presented, including a discussion on the mechanism, suppression methods, correction techniques, and utilization. Various methods for inhibition and calibration self-absorption effects are also categorized, compared and summarized, to provide ideas for improving the elemental detection accuracy of this technique.

For quantitative models and calibration algorithms, it is recommended to establish a library of models and calibration algorithms. Based on various experimental conditions, target elements, and spectral line parameter information, appropriate models and calibration algorithms can be selected to achieve rapid calibration of self-absorption effects. Furthermore, another important research direction is to explore and enhance the utilization of self-absorption effects to acquire plasma information and assisting in quantitative analysis.

Through in-depth research on the physical mechanisms, calibration methods, and utilization techniques of self-absorption effects, the control of self-absorption effects is expected to become more effective. LIBS technology, with its advantages of real-time analysis, rapid response, minimal sample damage, and simultaneous multi-element analysis, is poised to find more significant applications among various analytical techniques, potentially enabling its large-scale adoption.

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Notes

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REFERENCES

1. S. D. Sims. The effects of temperature on the output of a pulsed ruby laser. Ph.D., Purdue University, United States - Indiana, 1962.6300170. https://docs.lib.purdue.edu/dissertations/AAI6300170/
2. T. R. Loree and L. J. Radziemski, *Plasma Chem. Plasma Process.*, 1981, **1**, 271-279. https://doi.org/10.1007/BF00568835
3. F. Z. Xie, X. C. Lin, X. P. Wu, and Z. H. Xie, *Talanta.*, 2008, **74**, 836-843. https://doi.org/10.1016/j.talanta.2007.07.018
4. F. Zewdu and M. Amare, *Cogent Chem.*, 2018, **4**, 1534566. https://doi.org/10.1080/23312009.2018.1534566
5. V. Arancibia, M. Valderrama, K. Silva, and T. Tapia, *J. Chromatogr. B*, 2003, **785**, 303-309. https://doi.org/10.1016/s1570-0232(02)00924-8
6. Y. Tang, J. Li, Z. Hao, S. Tang, Z. Zhu, L. Guo, X. Li, X. Zeng, J. Duan, and Y. Lu, *Opt. Express*, 2018, **26**, 12121-12130. https://doi.org/10.1364/oe.26.012121
7. D. Wild, L. Pschyklenk, C. Theiss, and G. Holl, *Laser Technol. Defen. Secur.*, 2017, **10192**, 68-77. https://doi.org/10.1117/12.2263752
8. C. J. Zhao, D. M. Dong, X. F. Du, and W. G. Zheng, *Sensors*, 2016, **16**, 1764. https://doi.org/10.3390/s16101764
9. Q. Zeng, C. Y. Pan, C. Y. Li, T. Fei, X. K. Ding, X. W. Du, and Q. P. Wang, *Spectrochim. Acta B*, 2018, **142**, 68-73. https://doi.org/10.1016/j.sab.2018.01.011

10. L. Quackatz, A. Griesche, J. Nietzke, and T. Kannengiesser, *Weld. World*, 2024, **68**, 915-923. https://doi.org/10.1007/s40194-023-01677-2
11. R. Tosin, F. Monteiro-Silva, R. Martins, and M. Cunha, *Hortic.*, 2024, **10**, 82. https://doi.org/10.3390/horticulturae10010082
12. L. Ren, S. Li, W. Ye, Q. Lv, Y. Sun, X. Zhou, S. Lian, J. Lv, S. Wang, J. Guo, Y. Tian, R. Zheng, and Y. Lu, *Talanta.*, 2024, **271**, 125658. https://doi.org/10.1016/j.talanta.2024.125658
13. J. Li, L. Yao, L. Shen, and C. Wang, *Optoelectron. Lett.*, 2024, **20**, 116-121. https://doi.org/10.1007/s11801-024-3080-y
14. J. Li, L. Yao, L. Shen, and C. Wang, *Laser Phys.*, 2024, **34**, 045601. https://doi.org/10.1088/1555-6611/ad2443
15. M. Marković, D. Ranković, M. Savić, I. Perović, D. Milovanović, and M. Kuzmanović, *Meas. Sci. Technol.*, 2023, **35**, 1361-6501. https://doi.org/10.1088/1361-6501/ad060d
16. P. Baruah, A. Phukan, S. Sharma, and A. Nath, *Spectrochim. Acta, B*, 2023, **206**, 106726. https://doi.org/10.1016/j.sab.2023.106726
17. Z. Sun, C. Yu, J. Feng, J. Zhu, and Y. Liu, *J. Anal. At. Spectrom.*, 2024, **39**, 1212-1224. http://doi.org/10.1039/D4JA00002A
18. A. F. Shalabi, Y. S. Wudil, M. A. Al-Osta, O. S. B. Al-Amoudi, M. A. Gondal, and A. Almohammedi, *Appl. Spectrosc. Rev.*, 2024. https://doi.org/10.1080/05704928.2024.2302905
19. Z. J. Kamil, M. J. Zoory, and H. J. Mohamad, *Eur. Phys. J. D*, 2024, **78**, 1-10. https://doi.org/10.1140/epjd/s10053-024-00818-6
20. F. Zhou, Y. F. Liu, W. Y. Xie, J. Huang, F. Liu, W. W. Kong, Z. F. Zhao, and J. Y. Peng, *Crit. Rev. Food Sci. Nutr.*, 2023.1-17. https://doi.org/10.1080/10408398.2023.2283579
21. Z. Y. Peng, P. K. Yin, D. Li, Y. Y. Chen, K. Q. Shu, Q. W. Fan, Y. X. Duan, and Q. Y. Lin, *J. Innovative Opt. Health Sci.*, 2024, **17**, 2330008. https://doi.org/10.1142/S1793545823300082
22. V. H. C. Ferreira, V. Gardette, B. Busser, L. Sancey, S. Ronsmans, V. Bonneterre, V. Motto-Ros, and L. Duponchel, *Anal. Chem.*, 2024, **96**, 7038–7046. https://doi.org/10.1021/acs.analchem.4c00237
23. D. Zhang, X. Niu, J. Nie, S. Shi, H. Ma, and L. Guo, *Opt. Express*, 2024, **32**, 10851-10861. https://doi.org/10.1364/OE.515064
24. L. Duponchel, A. Cugerone, M. Muñoz, and V. Motto-Ros, *Spectrochim. Acta B*, 2023, **207**, 106758. https://doi.org/10.1016/j.sab.2023.106758
25. Y. Tang, S. X. Ma, R. Yuan, Y. Y. Ma, W. Sheng, S. P. Zhan, J. N. Wang, and L. B. Guo, *Opt. Lasers Eng.*, 2020, **134**, 106254. https://doi.org/10.1016/j.optlaseng.2020.106254
26. R. Zhou, K. Liu, Z. Y. Tang, P. Y. Gao, J. J. Yan, and X. Y. Li, *Appl. Opt.*, 2021, **60**, 9062-9066. https://doi.org/10.1364/AO.433538
27. T. Takahashi and B. Thornton, *Spectrochim. Acta B*, 2017, **138**, 31-42. https://doi.org/10.1016/j.sab.2017.09.010
28. H. C. Jin, X. S. Zhang, X. Wan, and J. W. Wang, *J. Soc. Leath. Tech. Ch.*, 2016, **100**, 182-185. https://webofscience.clarivate.cn/wos/woscc/full-record/WOS:000384393600003
29. K. A. Meier and S. Recknagel, *Fresen J Anal Chem*, 1997, **358**, 800-803. https://doi.org/10.1007/s002160050512
30. M. Reich, A. Deditius, S. Chryssoulis, J. W. Li, C. Q. Ma, M. A. Parada, F. Barra, and F. Mittermayr, *Geochim. Cosmochim. Acta*, 2013, **104**, 42-62. https://doi.org/10.1016/j.gca.2012.11.006
31. J.-M. Li, L.-B. Guo, C.-M. Li, N. Zhao, X.-Y. Yang, Z.-Q. Hao, X.-Y. Li, X.-Y. Zeng, and Y.-F. Lu, *Opt. Lett.*, 2015, **40**, 5224-5226. https://doi.org/10.1364/ol.40.005224
32. F. Deng, Z.-L. Hu, H.-h. Cui, D. Zhang, Y. Tang, Z.-F. Zhao, Q.-D. Zeng, and L.-B. Guo, *Spectrosc. Spect. Anal.*, 2021, **41**, 2989-2998. https://doi.org/10.3964/j.issn.1000-0593(2021)10-2989-10
33. F. Rezaei, G. Cristoforetti, E. Tognoni, S. Legnaioli, V. Palleschi, and A. Safi, *Spectrochim. Acta B*, 2020, **169**, 105878. https://doi.org/10.1016/j.sab.2020.105878
34. J. Hou, L. Zhang, Y. Zhao, Z. Wang, Y. Zhang, W. Ma, L. Dong, W. Yin, L. Xiao, and S. Jia, *Plasma Sci. Technol*, 2019, **21**, 034016. https://doi.org/10.1088/2058-6272/aaf875
35. M. Kimura and G. Nakamura, *JPN. J. Phys*, 1923, **2**, 53-59.
36. R. W. Wood, *The London, Edinburgh, and Dublin Philosophical Magazine and Journal of Science*, 2009, **2**, 876-880. https://doi.org/10.1080/14786442608564117
37. J. Hermann, C. Boulmer-Leborgne, and D. Hong, *J. Appl. Phys.*, 1998, **83**, 691-696. https://doi.org/10.1063/1.366639
38. A. M. El Sherbini, T. M. El Sherbini, H. Hegazy, G. Cristoforetti, S. Legnaioli, V. Palleschi, L. Pardini, A. Salvetti, and E. Tognoni, *Spectrochim. Acta B*, 2005, **60**, 1573-1579. https://doi.org/10.1016/j.sab.2005.10.011
39. M. C. D. Bulajic, G. Cristoforetti, S. Legnaioli, V. Palleschi, A. Salvetti, E. Tognoni, *Spectrochim. Acta B*, 2002, **57**, 339–353. https://doi.org/10.1016/S0584-8547(01)00398-6
40. A. B. Hssaine Amamoua, Belkacem Ferhatb, Roland Redona, Bruno Rossettoa, Pascal Matherona, *J. Quant. Spectrosc. Radiat. Transfer*, 2002,**75**, 747–763. https://doi.org/10.1016/S0022-4073(02)00040-7
41. H. O. Di Rocco, F. Bredice, and V. Palleschi, *Appl. Spectrosc*, 2011, **65**, 1213-1217. https://doi.org/10.1366/11-06335
42. B. Praher, V. Palleschi, R. Viskup, J. Heitz, and J. D. Pedarnig, *Spectrochim. Acta B*, 2010, **65**, 671-679. https://doi.org/10.1016/j.sab.2010.03.010
43. N. Konjević, M. Ivković, and S. Jovićević, *Spectrochim. Acta B*, 2010, **65**, 593-602. https://doi.org/10.1016/j.sab.2010.03.009
44. F. O. Bredice, H. O. Di Rocco, H. M. Sobral, M. Villagran-Muniz, and V. Palleschi, *Appl. Spectrosc.*, 2010, **64**, 320-323. https://doi.org/10.1366/000370210790918454
45. D. M. D. Pace, G. Bertuccelli, and C. A. D'Angelo, *J. Phys. Conf. Ser.*, 2011, **274**, 012076. https://doi.org/10.1088/1742-6596/274/1/012076
46. L. Gouy, *C.R.*, 1879, **88**, 418-421.
47. H.-Y. Moon, K. K. Herrera, N. Omenetto, B. W. Smith, and J. D. Winefordner, *Spectrochim. Acta B*, 2009, **64**, 702-713. http://doi.org/10.1016/j.sab.2009.06.011
48. N. KonjevicH, *Phys. Rep.*, 1999, **316**, 339-401. https://doi.org/10.1016/S0584-8547(99)00004-X
49. A. C. G. Mitchell, M. W. Zemansky, and S. P. Davis, *Am. J. Phys.*, 1972, **40**, 1187-1187. https://doi.org/10.1119/1.1986797
50. I. B. Gornushkin, J. M. Anzano, L. A. King, B. W. Smith, N. Omenetto, J. D. Winefordner, *Spectrochim. Acta B*, 1999, **54**, 491-503. http://doi.org/10.1016/S0584-8547(99)00004-X
51. C. Aragón, J. Bengoechea, and J. A. Aguilera, *Spectrochim. Acta B*, 2001, **56**, 619-628. https://doi.org/10.1016/S0584-8547(01)00172-0
52. J. A. Aguilera, J. Bengoechea, and C. Aragón, *Spectrochim. Acta B*, 2003, **58**, 221237. http://doi.org/10.1016/S0584-8547(02)00258-6
53. Y. Tang, S. Ma, Y. Chu, T. Wu, Y. Ma, Z. Hu, L. Guo, X. Zeng, J. Duan, and Y. Lu, *Opt. Express*, 2019, **27**, 4261-4270. https://doi.org/10.1364/oe.27.004261

54. T. Fei, C. Y. Pan, Q. Zeng, Q. P. Wang, and X. W. Du, *Plasma Sci. Technol*, 2018, **20**, 045503. https://doi.org/10.1088/2058-6272/aaaada

55. I. A. U. Medina, D. D. Carneiro, S. Rocha, E. E. Farias, F. O. Bredice, and V. Palleschi, *Appl. Spectrosc.*, 2021, **75**, 774-780. https://doi.org/10.1177/00037028211006764

56. S. A. M. Mansour, *Opt. Photonics. J*, 2015, **5**, 79-90. https://doi.org/10.4236/opj.2015.53007

57. F. Bredice, F. O. Borges, H. Sobral, M. Villagran-Muniz, H. O. Di Rocco, G. Cristoforetti, S. Legnaioli, V. Palleschi, L. Pardini, A. Salvetti, and E. Tognoni, *Spectrochim. Acta B*, 2006, **61**, 1294-1303. https://doi.org/10.1016/j.sab.2006.10.015

58. D. M. D. Pace, C. A. D'Angelo, and G. Bertuccelli, *IEEE Trans. Plasma Sci.*, 2012, **40**, 898-908. http://doi.org/10.1109/Tps.2011.2181875

59. D. M. D. Pace, C. A. D'Angelo, and G. Bertuccelli, *Appl Spectrosc*, 2011, **65**, 1202-1212. https://doi.org/10.1366/11-06348

60. Y. Zhang, Y. Lu, Y. Tian, Y. Li, W. Ye, J. Guo, and R. Zheng, *Anal. Chim. Acta*, 2022, **1195**, 339423. https://doi.org/10.1016/j.aca.2021.339423

61. M. Burger, M. Skocic, and S. Bukvic, *Spectrochim. Acta B*, 2014, **101**, 51-56. https://doi.org/10.1016/j.sab.2014.07.007

62. M. Cvejic, M. R. Gavrilovic, S. Jovicevic, and N. Konjevic, *Spectrochim. Acta B*, 2013, **85**, 20-33. https://doi.org/10.1016/j.sab.2013.03.011

63. C. G. Parigger, D. M. Surmick, and G. Gautam, *J. Phys.: Conf. Ser.* , 2017, **810**, 0120121. https://doi.org/10.1088/1742-6596/810/1/012012

64. H. W. Drawin and F. Emard, *Beiträge aus der Plasmaphysik*, 1973, **13**, 143-168. https://doi.org/10.1002/ctpp.19730130304

65. F. E. Irons, *J. Quant. Spectrosc. Radiat. Transfer*, 1979, **22**, 1-20. https://doi.org/10.1016/0022-4073(79)90102-X

66. F. E. Irons, *J. Quant. Spectrosc. Radiat. Transfer*, 1979, **22**, 21-36. https://doi.org/10.1016/0022-4073(79)90103-1

67. F. E. Irons, *J. Quant. Spectrosc. Radiat. Transfer*, 1979, **22**, 37-44. https://doi.org/10.1016/0022-4073(79)90104-3

68. F. E. Irons, *J. Quant. Spectrosc. Radiat. Transfer*, 1980, **24**, 119-132. https://doi.org/10.1016/0022-4073(80)90025-4

69. R. Hannachi, S. Boussaïdi, P. Teulet, G. Taieb, Y. Cressault, A. Gleizes, and Z. B. Lakhdar, *Appl. Phys. A*, 2008, **92**, 933-937. https://doi.org/10.1007/s00339-008-4585-0

70. R. Hannachi, Y. Cressault, Ph. Teulet, A. Gleizes, and Z. Ben Lakhdar, *Spectrochim. Acta B*, 2008, 63, 1054-1059. https://doi.org/10.1016/j.sab.2008.08.010

71. J. He and L. Xia, *Mod. Phys. Lett. B*, 2010, **24**, 2267-2275. https://doi.org/10.1142/s0217984910024560

72. V. Lazic, R. Barbini, F. Colao, R. Fantoni, and A. Palucci, *Spectrochim. Acta B*, 2001, **56**, 807-820. https://doi.org/10.1016/s0584-8547(01)00211-7

73. P. B. Hansen, S. Schroeder, S. Kubitza, K. Rammelkamp, D. S. Vogt, and H. W. Huebers, *Spectrochim. Acta B*, 2021, **178**, 106115. https://doi.org/10.1016/j.sab.2021.106115

74. J. Ben Ahmed and J. Cowpe, *Appl. Opt.*, 2010, **49**, 3607-3612. http://doi.org/10.1364/Ao.49.003607

75. L. X. Sun and H. B. Yu, *Talanta.*, 2009, **79**, 388-395. https://doi.org/10.1016/j.talanta.2009.03.066

76. H. Shakeel, S. U. Haq, G. Aisha, and A. Nadeem, *Phys. Plasma*, 2017, **24**, 063516. https://doi.org/10.1063/1.4985327

77. G. Zhang, H. Song, Y. Jin, J. Ma, S. Li, and Z. Ren, *Optik*, 2019, **183**, 375-380. https://doi.org/10.1016/j.ijleo.2019.02.082

78. Z. Ramezanian, S. M. R. Darbani, and A. E. Majd, *Appl. Opt.*, 2017, **56**, 6917-6922. https://doi.org/10.1364/Ao.56.006917

79. Z. Farooq, R. Ali, A. Ul Ahmad, M. Yaseen, M. H. R. Mahmood, M. Fahad, M. N. Hussain, I. Rehan, M. Z. Khan, Ramiza, M. U. Farooq, M. A. Qayyum, and M. Shafique, *Appl. Opt.*, 2020, **59**, 2559-2568. http://doi.org/10.1364/ao.379641

80. S. M. Z. Iqbal, Z. Uddin, Z. A. Umar, N. Ahmed, R. Ahmed, and M. A. Baig, *Anal. Lett.*, 2022, **55**, 11-23. https://doi.org/10.1080/00032719.2021.1910831

81. Z. Farooq, R. Ali, A. Ali, T. Mubeen, T. Jan, and H. Anwar, *Appl. Spectrosc.*, 2019, **73**, 30-39. https://doi.org/10.1177/0003702818789959

82. F. d. O. Borges, J. U. Ospina, G. d. H. Cavalcanti, E. E. Farias, A. A. Rocha, P. I. L. B. Ferreira, G. C. Gomes, and A. Mello, *J. Anal. At. Spectrom.*, 2018, **33**, 629-641. https://doi.org/10.1039/c7ja00299h

83. H. Fu, F. Dong, H. Wang, J. Jia, and Z. Ni, *Appl. Spectrosc.*, 2017, **71**, 1982-1989. https://doi.org/10.1177/0003702817699410

84. T. Li, Z. Hou, Y. Fu, J. Yu, W. Gu, and Z. Wang, *Anal. Chim. Acta*, 2019, **1058**, 39-47. https://doi.org/10.1016/j.aca.2019.01.016

85. G. H. Cavalcanti, D. V. Teixeira, S. Legnaioli, G. Lorenzetti, L. Pardini, and V. Palleschi, *Spectrochim. Acta B*, 2013, **87**, 51-56. https://doi.org/10.1016/j.sab.2013.05.016

86. H. Fu, H. Wang, J. Jia, Z. Ni, and F. Dong, *Appl. Spectrosc.*, 2018, **72**, 1183-1188. https://doi.org/10.1177/0003702818770572

87. A. N. Kadachi, M. A. Al-Eshaikh, and K. Ahmad, *Laser Phys.*, 2018, **28**, 095710. http://doi.org/10.1088/1555-6611/aac9e5

88. F. Deng, Z. Hu, D. Zhang, F. Chen, X. Niu, J. Nie, Q. Zeng, and L. Guo, *Opt. Express*, 2022, **30**, 9256-9268. https://doi.org/10.1364/oe.446334

89. Z. Hu, J. Nie, Z. Ouyang, D. Zhang, Y. Liu, Y. Chu, and L. Guo, *Opt. Lett.*, 2023, **48**, 1-4. http://doi.org/10.1364/OL.472224

90. D. Díaz, A. Molina, and D. Hahn, *Spectrochim. Acta, Part B*, 2018, **145**, 86-95. https://doi.org/10.1016/j.sab.2018.04.007

91. H. I. G. Abdellatifa, *Spectrochim. Acta B*, 2002, **57**, 1155-1165. https://doi.org/10.1016/S0584-8547(02)00057-5

92. H. Cui, Y. Tang, S. Ma, Y. Ma, D. Zhang, Z. Hu, Z. Wang, and L. Guo, *Optik*, 2020, **204**, 164144. https://doi.org/10.1016/j.ijleo.2019.164144

93. Y. I. Lee, K. Song, H. K. Cha, J. M. Lee, M. C. Park, G. H. Lee, and J. Sneddon, *Appl. Spectrosc.*, 1997, **51**, 959-964. https://doi.org/10.1366/0003702971941610

94. N. Li, K. Tanabe, N. Nishi, R. Zheng, and T. Sakka, *J. Anal. At. Spectrom.*, 2021, **36**, 1960-1968. http://doi.org/10.1039/d1ja00151e

95. N. M. Sabri, Z. Haider, K. Tufail, F. D. Ismail, and J. Ali, *Phys. Plasma*, 2018, **25**, 073303. https://doi.org/10.1063/1.5023666

96. M. Kuzuya and H. Aranami, *Spectrochim. Acta B*, 2000, **55**, 1423-1430. https://doi.org/ 10.1016/S0584-8547(00)00246-9

97. M. Pardede, I. Karnadi, Z. S. Lie, E. Jobiliong, I. Tanra, R. Hedwig, A. M. Marpaung, H. Suyanto, M. M. Suliyanti, D. Kwaria, T. J. Lie, K. H. Kurniawan, and K. Kagawa, *Opt. Express*, 2021, **29**, 22593-22602. https://doi.org/10.1364/oe.431784

98. A. S. Altowyan, A. El-Hussein, H. A. Ahmed, H. H. Faqih, A. S. Amein, A. M. Mostafa, and M. M. ElFaham, *Opt. Mater.*, 2022, **131**, 112731. https://doi.org/10.1016/j.optmat.2022.112731

99. W. Ke, H. Yuan, J. Q. Liu, X. H. Wang, A. J. Yang, J. F. Chu, D. X. Liu, and M. Z. Rong, *J. Phys. D: Appl. Phys.*, 2024, **57**, 095204. https://doi.org/10.1088/1361-6463/ad0e98

100. M. Pardede, I. Karnadi, Z. S. Lie, E. Jobiliong, I. Tanra, R. Hedwig, A. M. Marpaung, H. Suyanto, M. M. Suliyanti, D. Kwaria, T. J. Lie, K. H. Kurniawan, and K. Kagawa, *Opt Express*, 2021, **29**, 22593-22602. https://doi.org/10.1364/OE.431784

101. R. Hai, Z. He, X. Yu, L. Sun, D. Wu, and H. Ding, *Opt. Express*, 2019, **27**, 2509-2520. https://doi.org/10.1364/oe.27.002509

102. R. Hedwig, I. Tanra, I. Karnadi, M. Pardede, A. M. Marpaung, Z. S. Lie, K. H. Kurniawan, M. M. Suliyanti, T. J. Lie, and K. Kagawa, *Opt. Express*, 2020, **28**, 9259-9268. https://doi.org/10.1364/oe.387946

103. J. Wang, Z. Liu, L. Zhu, Z. Song, Y. Zhang, L. Zhang, W. Zhang, G. Wang, Z. Ye, Z. Zhu, W. Yin, and S. Jia, *Opt Express*, 2023, **31**, 16423-16433. https://doi.org/10.1364/OE.489720

104. M. M. Suliyanti, S. Sardy, A. Kusnowo, R. Hedwig, S. N. Abdulmadjid, K. H. Kurniawan, T. J. Lie, M. Pardede, K. Kagawa, and M. O. Tjia, *J. Appl. Phys.*, 2005, **97**, 053305-053309. https://doi.org/10.1063/1.1858061

105. J. F. Y. Gravel and D. Boudreau, *Spectrochim. Acta B*, 2009, **64**, 56-66. https://doi.org/10.1016/j.sab.2008.10.037

106. Q. L. Ma, V. Motto-Ros, W. Q. Lei, M. Boueri, X. S. Bai, L. J. Zheng, H. P. Zeng, and J. Yu, *Spectrochim. Acta B*, 2010, **65**, 896-907. https://doi.org/10.1016/j.sab.2010.08.005

107. Z. Q. Hao, L. Liu, M. Shen, X. Y. Yang, K. H. Li, L. B. Guo, X. Y. Li, Y. F. Lu, and X. Y. Zeng, *Opt. Express*, 2016, **24**, 26521-26528. https://doi.org/10.1364/oe.24.026521

108. Y. Zhang, Y. Tian, Y. Lu, L. Guo, Y. Li, J. Guo, and R. Zheng, *J. Anal. At. Spectrom.*, 2021, **36**, 644-653. https://doi.org/10.1039/d0ja00451k

109. B. Yuan, R.-B. Ning, Q. Li, Y.-L. Han, and S.-N. Xu, *Spectrosc. Spect. Anal.*, 2020, **40**, 2891-2895. https://doi.org/10.3964/j.issn.1000-0593(2020)09-2891-05

110. R. Yi, X. Yang, F. Lin, and S. Ren, *Appl. Opt.*, 2019, **58**, 8824-8828. https://doi.org/10.1364/ao.58.008824

111. R.-B. Ning, C.-X. Li, Q. Li, B. Yuan, and S.-N. Xu, *Spectrosc. Spect. Anal.*, 2018, **38**, 3546-3549. https://doi.org/10.3964/j.issn.1000-0593(2018)11-3546-04

112. R. Brennetot, J. L. Lacour, E. Vors, A. Rivoallan, D. Vailhen, and S. Maurice, *Appl. Spectrosc*, 2003, **57**, 744-752. https://doi.org/10.1366/000370203322102816

113. C. Y. Diao, C. S. Chen, B. Y. Man, C. Wang, and H. B. Fu, *Eur. Phys. J. D*, 2011, **63**, 123-128. https://doi.org/10.1140/epjd/e2011-10394-y

114. H. B. Fu, C. S. Chen, B. Y. Man, X. Song, and D. Liu, *Eur. Phys. J. D*, 2011, **65**, 509-513. https://doi.org/10.1140/epjd/e2011-20265-2

115. A. M. Marpaung, E. Harefa, M. Pardede, I. Karnadi, R. Hedwig, I. Tanra, M. M. Suliyanti, Z. S. Lie, M. Shiddiq, M. Ramli, K. Lahna, E. Jobiliong, S. N. Abdulmadjid, N. Idris, A. Khumaeni, W. Setiabudi, H. Suyanto, T. J. Lie, K. H. Kurniawan, and K. Kagawa, *Heliyon*, 2022, **8**, e10057. https://doi.org/10.1016/j.heliyon.2022.e10057

116. R. Yi, L. Guo, C. Li, X. Yang, J. Li, X. Li, X. Zeng, and Y. Lu, *J. Anal. At. Spectrom.*, 2016, **31**, 961-967. https://doi.org/10.1039/c5ja00500k

117. I. Urbina, D. Carneiro, S. Rocha, E. E. Farias, F. Bredice, and V. Palleschi, *Spectrochim. Acta B*, 2020, **170**, 105902. https://doi.org/10.1016/j.sab.2020.105902

118. J. Hou, L. Zhang, W. Yin, S. Yao, Y. Zhao, W. Ma, L. Dong, L. Xiao, and S. Jia, *Opt. Express*, 2017, **25**, 23024-23034. https://doi.org/10.1364/oe.25.023024

119. J. Ben Ahmed and F. Fouad, *Ieee Trans Plasma Sci.*, 2014, **42**, 2073-2078. https://doi.org/10.1109/tps.2014.2331851

120. G.-C. He, X.-L. Zhu, L.-N. Shi, S.-J. Zhao, and Y.-L. Hua, *Opt. Laser Technol.*, 2021, **143**, 107324. https://doi.org/10.1016/j.optlastec.2021.107324

121. G. C. Y. Chan, L. R. Martin, L. D. Trowbridge, Z. Zhu, X. Mao, and R. E. Russo, *Spectrochim. Acta B*, 2021, **176**, 106036. https://doi.org/10.1016/j.sab.2020.106036

122. M. Hornackova, J. Plavcan, M. Hornacek, P. Hudec, and P. Veis, *Atoms*, 2019, **7**, 98. https://doi.org/10.3390/atoms7040098

123. J. A. Aguilera and C. Aragón, *Spectrochim. Acta, Part B*, 2008, **63**, 793-799. https://doi.org/10.1016/j.sab.2008.04.013

124. L. M. John and K. K. Anoop, *RSC Adv*, 2023, **13**, 29613-29624. https://doi.org/10.1039/d3ra06226k

125. J. Viljanen, Z. W. Sun, and Z. T. Alwahabi, *Spectrochim. Acta B*, 2016, **118**, 29-36. https://doi.org/10.1016/j.sab.2016.02.002

126. Y. Fan, Y. Gu, Z. Hu, F. Chen, J. Nie, Y. Liu, W. Cheng, and L. Guo, *J. Anal. At. Spectrom.*, 2023, **38**, 1713-1719. https://doi.org/10.1039/d3ja00098b

127. J. M. Li, L. B. Guo, C. M. Li, N. Zhao, X. Y. Yang, Z. Q. Hao, X. Y. Li, X. Y. Zeng, and Y. F. Lu, *Opt. Lett.*, 2015, **40**, 5224-5226. https://doi.org/10.1364/Ol.40.005224

128. J. Li, Y. Tang, Z. Hao, N. Zhao, X. Yang, H. Yu, L. Guo, X. Li, X. Zeng, and Y. Lu, *J. Anal. At. Spectrom.*, 2017, **32**, 2189-2193. https://doi.org/10.1039/c7ja00199a

129. Y. Tang, L. Guo, J. Li, S. Tang, Z. Zhu, S. Ma, X. Li, X. Zeng, J. Duan, and Y. Lu, *J. Anal. At. Spectrom.*, 2018, **33**, 1683-1688. https://doi.org/10.1039/c8ja00147b

130. Y. Tang, S. Ma, R. Yuan, Y. Ma, W. Sheng, S. Zhan, J. Wang, and L. Guo, *Opt. Lasers Eng.*, 2020, **134**, 106254. https://doi.org/10.1016/j.optlaseng.2020.106254

131. S. L. Chen, R. H. Li, and Y. Q. Chen, *Appl. Opt.*, 2023, **62**, 4512-4517. https://doi.org/10.1364/AO.492394

132. Z. Xiong, Z. Hao, X. Li, and X. Zeng, *J. Anal. At. Spectrom.*, 2019, **34**, 1606-1610. https://doi.org/10.1039/c9ja00078j

133. J. J. Hou, L. Zhang, Y. Zhao, W. G. Ma, L. Dong, W. B. Yin, L. T. Xiao, and S. T. Jia, *Opt. Express*, 2019, **27**, 32184-32192. https://doi.org/10.1364/oe.27.032184

134. J.-J. Hou, L. Zhang, Y. Zhao, W.-B. Yin, L. Dong, W.-G. Ma, L.-T. Xiao, and S.-T. Jia, *Spectrosc.Spect. Anal.*, 2020, **40**, 261-265. http://doi.org/10.3964/j.issn.1000-0593(2020)01-0261-05

135. J. Li, J. Hou, W. Zhang, Y. Zhao, F. Chen, S. Wang, L. Zhang, G. Wang, X. Ma, Z. Liu, X. Luo, W. Yin, and S. Jia, *Opt. Lasers Eng.*, 2023, **160**, 107231. https://doi.org/10.1016/j.optlaseng.2022.107231

136. I. Karnadi, M. Pardede, I. Tanra, R. Hedwig, A. M. Marpaung, Z. S. Lie, E. Jobiliong, D. Kwaria, M. M. Suliyanti, M. Ramli, K. Lahna, T. J. Lie, H. Suyanto, K. H. Kurniawan, and K. Kagawa, *Sci. Rep.*, 2020, 13278. https://doi.org/10.10.1038/s41598-020-70151-6

137. G. Quintana-Silva, H. Sobral, and J. Rangel-Cárdenas, *Chemosensors*, 2022, **11**, 4. https://doi.org/10.3390/chemosensors11010004

138. W. Wang, L. Sun, P. Zhang, L. Zheng, and L. Qi, *Microchem. J.*, 2022, **172**, 106964. http://doi.org/10.1016/j.microc.2021.106964

139. J. Li, Z. Tang, Z. Hao, Z. Huang, C. Zhu, Q. Li, B. He, J. Chen, and X. Li, *J. Anal. At. Spectrom.*, 2022, **37**, 2176-2181. https://doi.org/10.1039/d2ja00206j

140. L. Zhang, Y. Sun, J. Hou, W. Yin, L. Dong, W. Ma, L. Xiao, and S. Jia, *Sci. Sin. Phys.Mech. Astron.*, 2017, **47**, 124201. https://doi.org/10.1360/sspma2017-00222

141. X. Liu, Q. Zhang, Q. Xiong, and T. C. Sum, *Nano Lett.*, 2013, **13**, 1080-1085. https://doi.org/10.1021/nl304362u

142. A. M. El Sherbini, T. M. El Sherbini, H. Hegazy, G. Cristoforetti, S. Legnaioli, V. Palleschi, L. Pardini, A. Salvetti, and E. Tognoni, *Spectrochim. Acta B*, 2005, **60**, 1573-1579. https://doi.org/10.1016/j.sab.2005.10.011

143. G. Cristoforetti and E. Tognoni, *Spectrochim. Acta B*, 2013, **79-80**, 63-71. http://doi.org/10.1016/j.sab.2012.11.010

144. A. Safi, S. H. Tavassoli, G. Cristoforetti, S. Legnaioli, V. Palleschi, F. Rezaei, and E. Tognoni, *J. Adv. Res.*, 2019, **18**, 1-7. https://doi.org/10.1016/j.jare.2019.01.008

145. C. Aragón and J. A. Aguilera, *J. Quant. Spectrosc. Radiat. Transfer*, 2014, **149**, 90-102. https://doi.org/10.1016/j.jqsrt.2014.07.026

146. J. A. Aguilera, C. Aragón, and J. Manrique, *J. Quant. Spectrosc. Radiat. Transfer*, 2015, **160**, 10-18. https://doi.org/10.1016/j.jqsrt.2015.03.008

147. J. Manrique, J. A. Aguilera, and C. Aragón, *Mon. Not. R. Astron. Soc.*, 2018, **477**, 1922-1926. https://doi.org/10.1093/mnras/sty698

148. J. Manrique, D. M. D. Pace, C. Aragón, and J. A. Aguilera, *Mon. Not. R. Astron. Soc.*, 2020, **498**, 2068-2074. https://doi.org/10.1093/mnras/staa2000

149. J. A. Aguilera and C. Aragón, *J. Anal. At. Spectrom.*, 2017, **32**, 144-152. https://doi.org/10.1039/c6ja00360e

150. A. Safi, S. H. Tavassoli, G. Cristoforetti, E. Tognoni, B. Campanella, S. Legnaioli, S. Pagnotta, F. Poggialini, and V. Palleschi, *Anal. Chem.*, 2019, **91**, 8595-8601. https://doi.org/10.1021/acs.analchem.9b01885

151. F. Poggialini, B. Campanella, R. Jafer, S. Legnaioli, F. Bredice, S. Raneri, and V. Palleschi, *Spectrochim. Acta B*, 2020, **167**, 1905829. https:/doi.org/10.1016/j.sab.2020.105829

152. A. Safi, S. M. Aberkane, A. Botto, B. Campanella, S. Legnaioli, F. Poggialini, S. Raneri, F. Rezaei, and V. Palleschi, *Appl. Spectrosc.*, 2021, **75**, 654-660. https://doi.org/10.1177/0003702821999425

153. G.-C. He, L.-N. Shi, Y.-L. Hua, J.-B. Niu, C. Lu, and C.-Q. Xie, *Opt. Express*, 2020, **28**, 6018-6026. https://doi.org/10.1364/oe.382868

154. J. A. Aguilera, J. Bengoechea, and C. Aragón, *Spectrochim. Acta, B*, 2003, **58**, 221-237. https://doi.org/10.1016/S0584-8547(02)00258-6

155. Z. Hu, F. Chen, D. Zhang, Y. Chu, W. Wang, Y. Tang, and L. Guo, *Anal. Chim. Acta*, 2021, **1183**, 339008. https://doi.org/10.1016/j.aca.2021.339008